

ViTrox Corporation Berhad

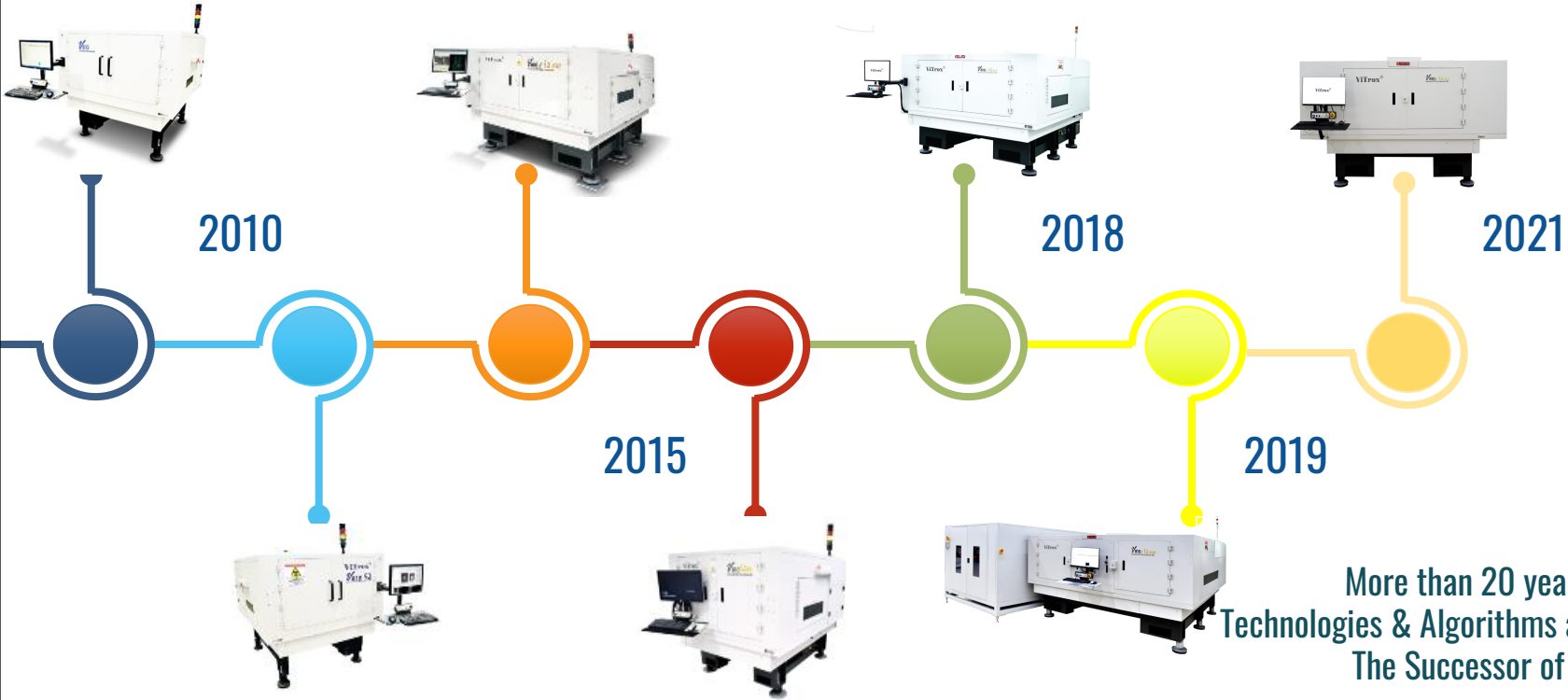
Introduction of Advanced 3D X-ray Inspection System (V810i)

www.vitrox.com



Vitrox Product Detail

Machine Development Time Stamp



More than 20 years of Recognized
Technologies & Algorithms at V810i Series :
The Successor of 5DX and X6000

Ever-Increasing Performance & Capability

Speed improvement % depends on board
complexity and size

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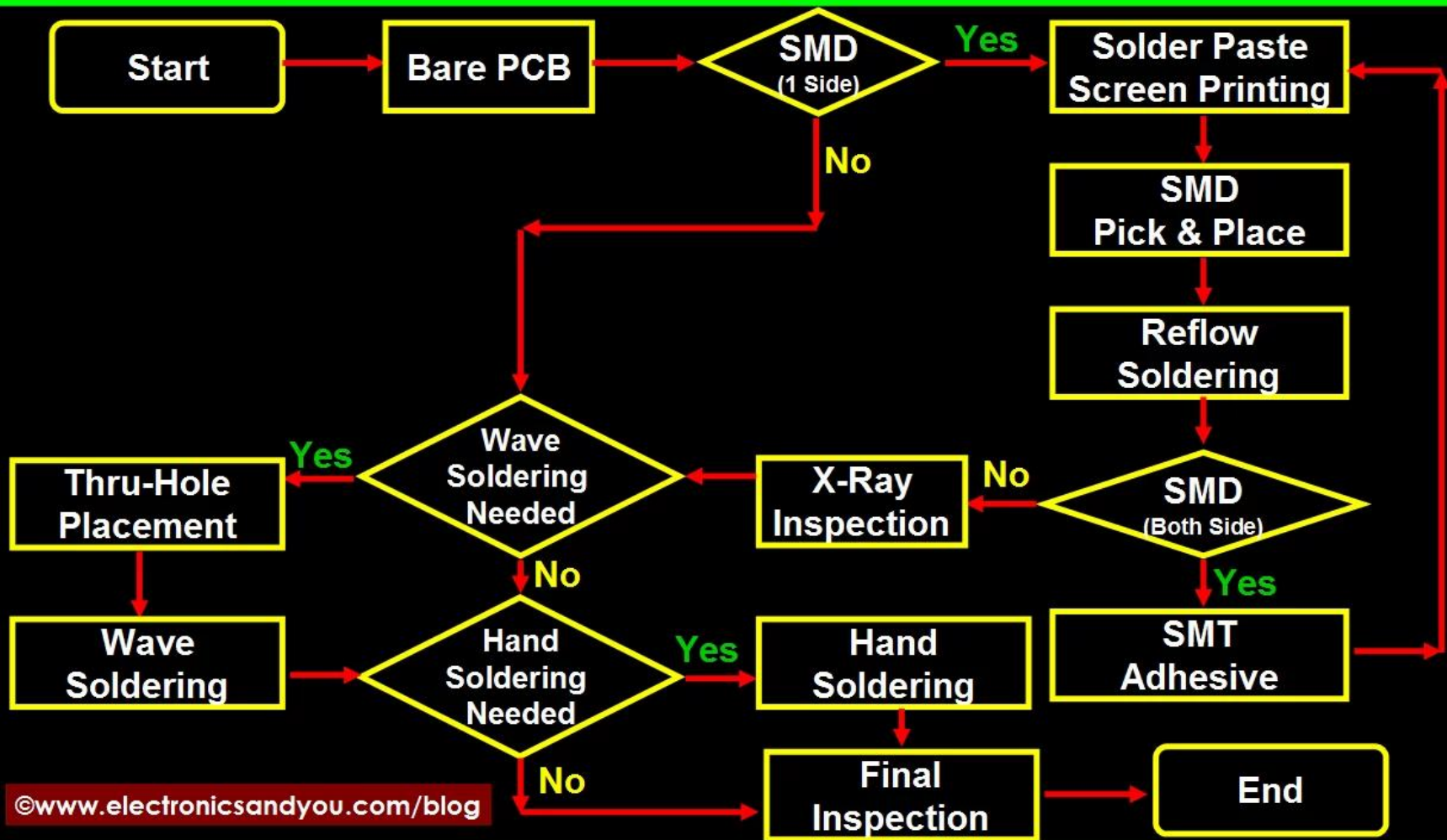
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Lesson 1

Advanced 3D X-ray Inspection System (AXI)

Sample of SMT Production Floor Plan





What is an X-ray Machine

An X-ray machine is the machine that involves x-rays. It usually consist of an x-ray generator, and an x-ray detector.

X-ray machine is widely used in different field:

- Medical - Projectional radiography (conventional radiography), computed tomography (CT Scan), Radiation therapy (not imaging).
- Airport security
- Revealing counterfeit art
- PCBA Industry.

Introduction of AXI

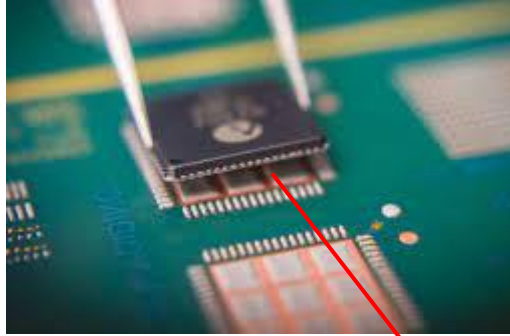
Automatic optical inspection (AOI) works very well in electronics manufacturing for printed circuit boards when joints are visible. However many PCBs today are using technologies such as ball grid array, BGA integrated circuits and chip scale packages, CSPs where the solder connections are not visible.

This has arisen as a result of the need for greater numbers of interconnections to integrated circuit packages and as a general result of increasing complexity. In these and many other instances it is necessary to carry out checks using automated X-Ray inspection (AXI) equipment that can check the solder joints under the components, and also reveal many defects in solder joints that may not be visible with ordinary optical inspection equipment.

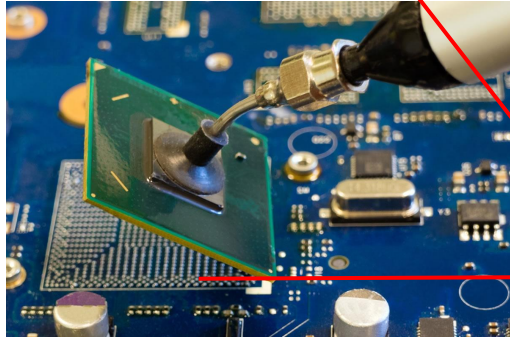
Introduction of AXI

- X-ray beam passes through the printed circuit board.
- Dense metal absorb more of the X-ray's energy than less dense material.
- The X-ray beam reaches the "Detector" (X-ray Camera).
- The "Detector" converts the X-ray energy into a gray scale image of a solder joint.
- The "Detector" then sends the gray scale image to the Imaging Reconstruction Processor (IRP).
- IRP collect the images, reconstruct the images to 3D Reconstructed Images and send the images to the computer.
- Computer identifies and measures specific pieces of the joint's image.

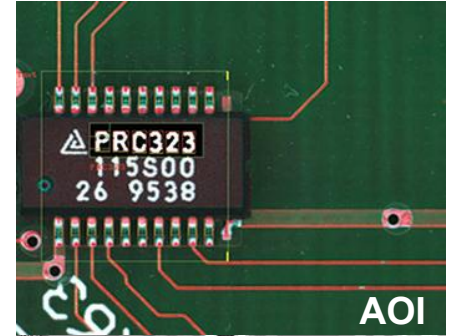
X-ray Inspection



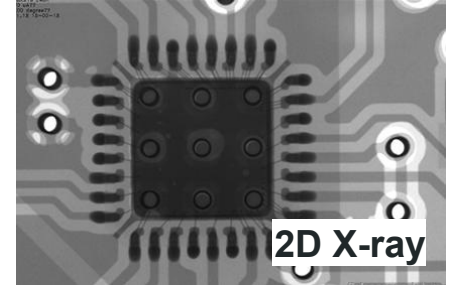
X-ray Inspection in SMT line is referring to components underneath inspection which are not visible from the surface. With x-ray penetration capability, defects are visible in the image for all type of packages.



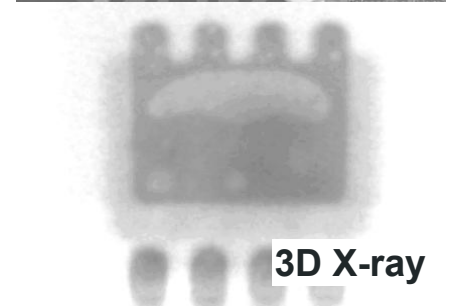
Hidden
Joint



AOI



2D X-ray



3D X-ray

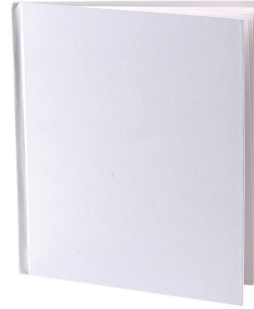
3D Automated X-ray Inspection

3D X-ray Inspection is referring to the ability of the system to slice through different z-height with clear separation of top and bottom component without any artifact.

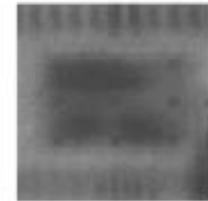
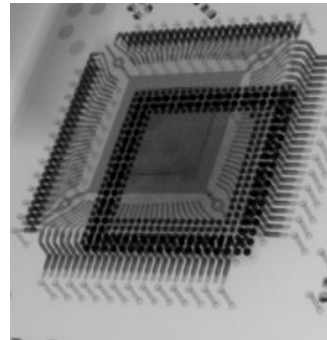
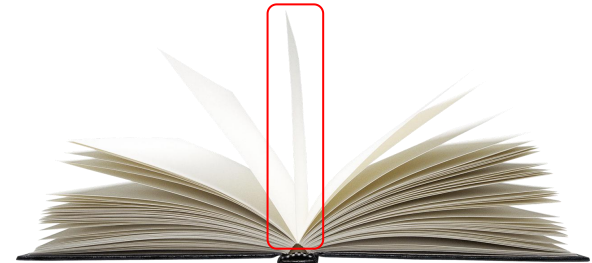
Comparison between 2D X-ray Inspection image and 3D x-ray Inspection image

- Ability of 3D system slice through different z-height
- Clear separation for top and bottom component

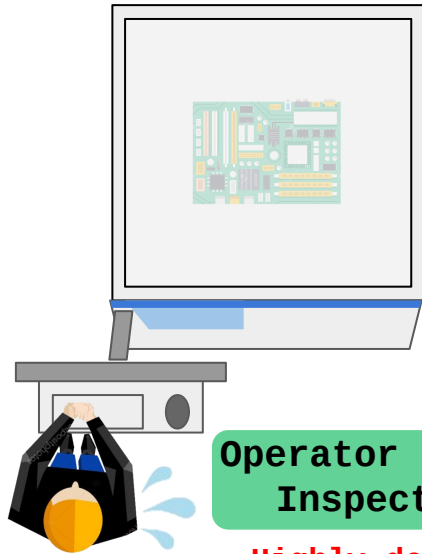
2D X-ray



3D X-ray



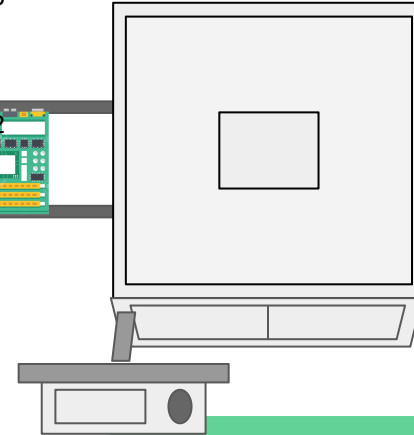
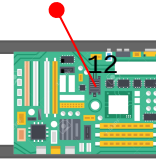
Automated Inspection



Operator Manual Inspection

Highly depends on user experience and only able to view hidden joint through 2D x-ray manually.

Board Size : 16"
x 14"
Joint Count : 27,000
BGA Components



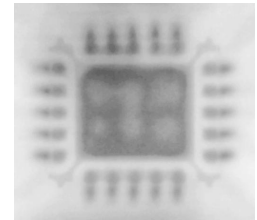
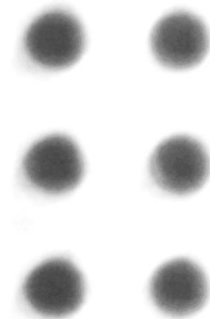
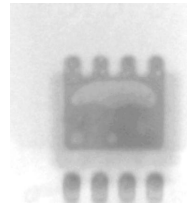
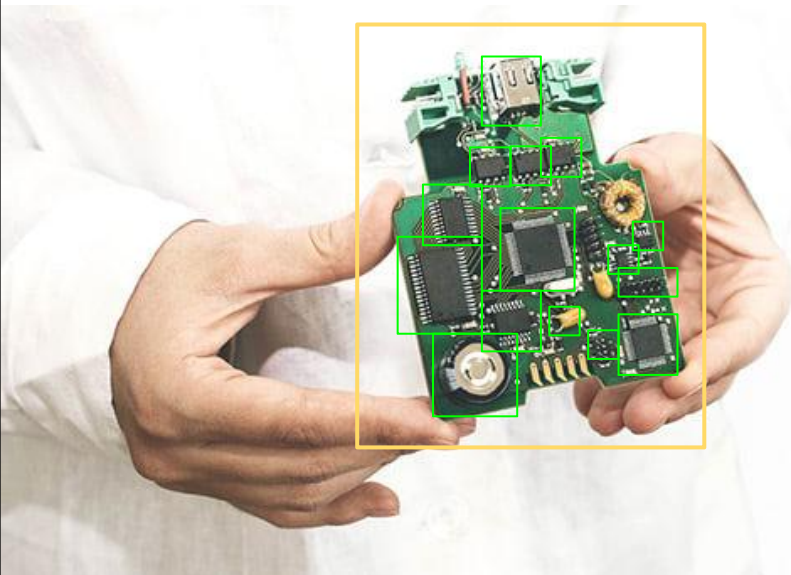
3D AXI Inspection

Able to complete the inspection within 90s for 100% test coverage

Automated Full Board Inspection

Why inspect only hidden joint while you can inspect full board?

X-ray is the best inspection method for solder joint context. Some defects like voiding in chip components are not detectable with AOI. And also dual inspection can provide better guarantee on board quality.

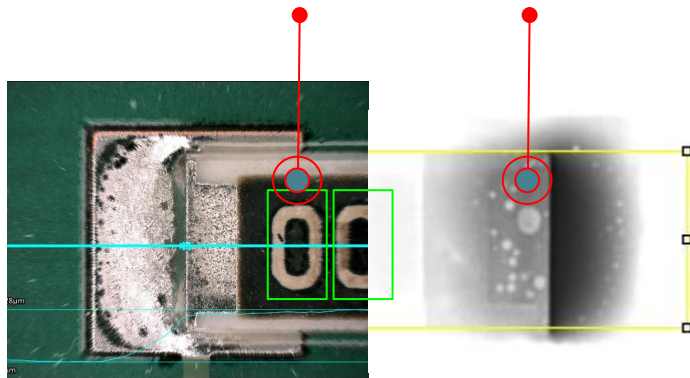


Automated Full Board Inspection

Why inspect only hidden joint while you can inspect full board?

OCV inspection(AOI)

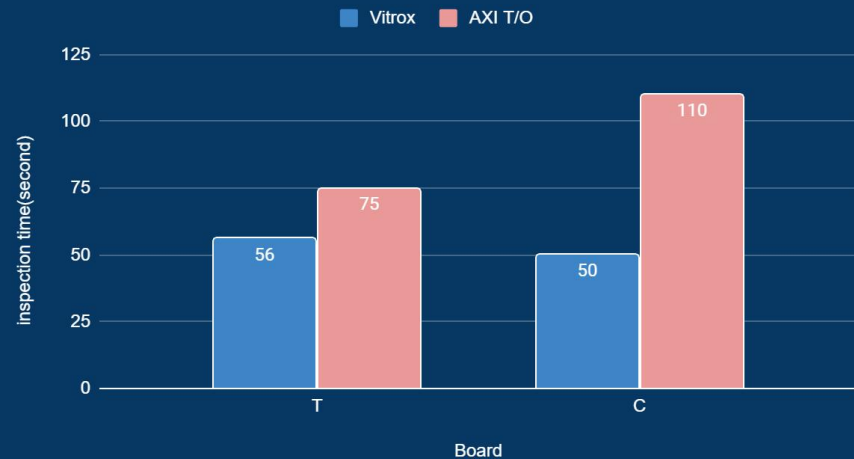
Void (AXI)



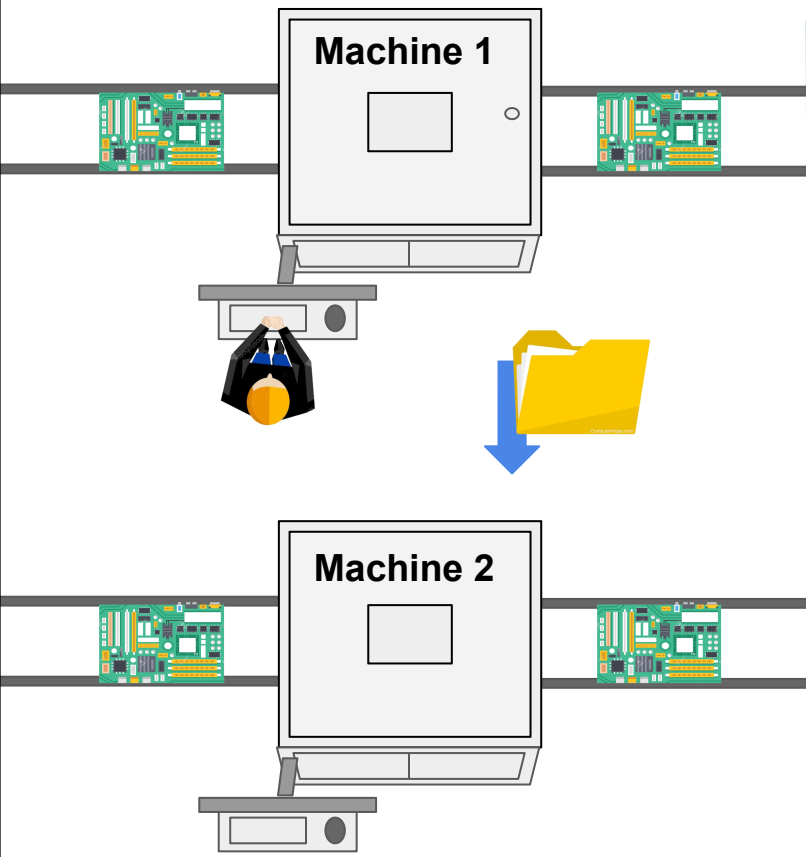
If there is a connection using solder paste, there is a chance that defect happen under the joint such as void where detection through surface is not possible.

ViTrox AXI able to achieve faster inspection cycle time compared to other type of AXI machine in the market

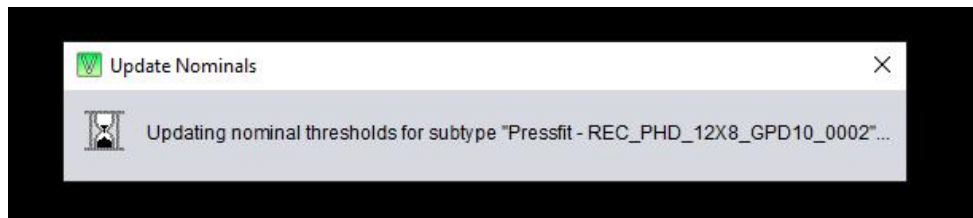
Full Board Inspection between ViTrox AXI and AXI T&O



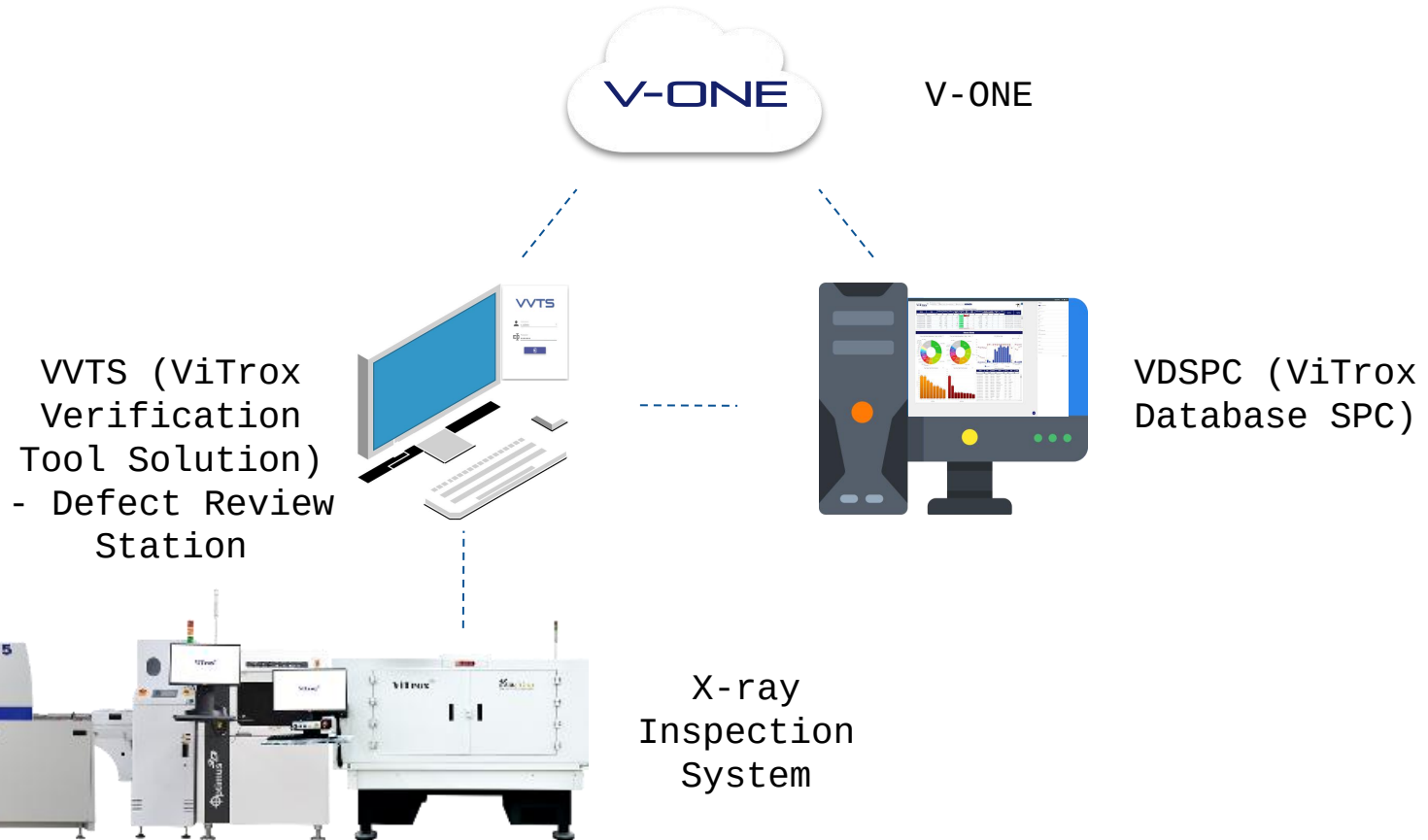
Program Transferability



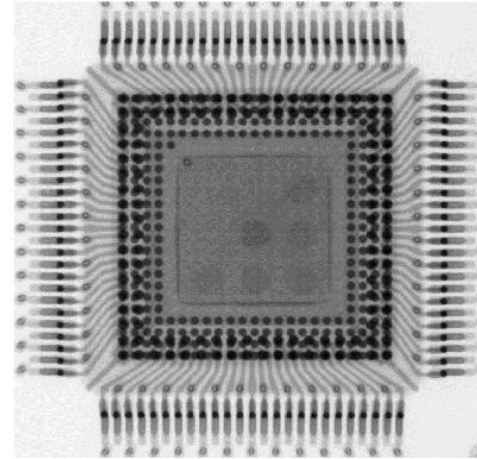
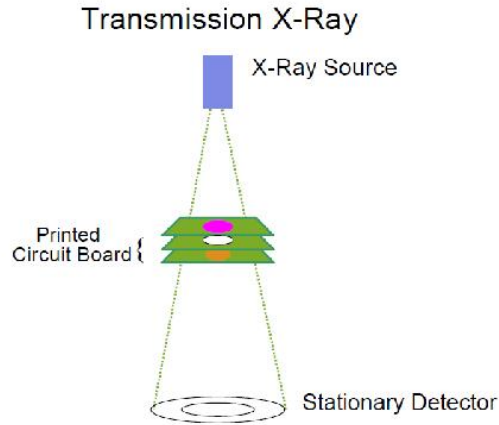
Similar program from same series of machine type to another will only required update program joint nominal value. Once complete, it is good to proceed for inspection.



Work Stations



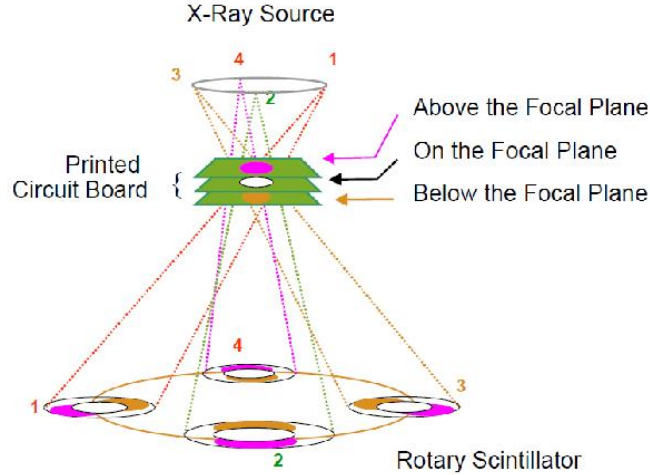
Transmission X-Ray (2D Image)



Transmission X-ray:

- The X-ray mean that most people are familiar with is the kind used by a doctor or dentist. These are “transmission” X-ray and regardless of the layering of the objects being x-ray produced an image of all features.

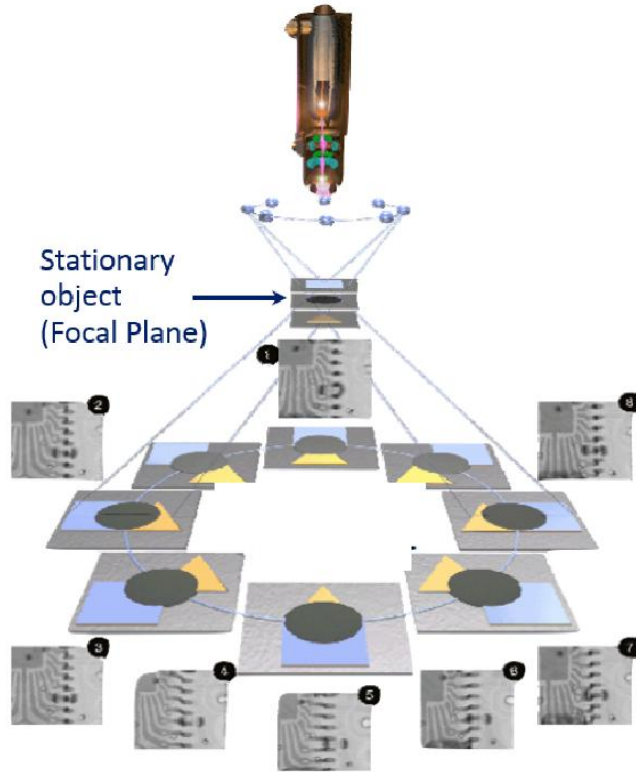
Laminography (5DX)



Laminography X-ray Beam:

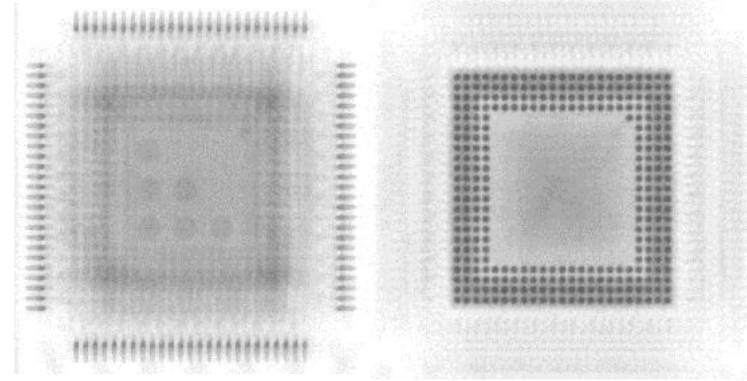
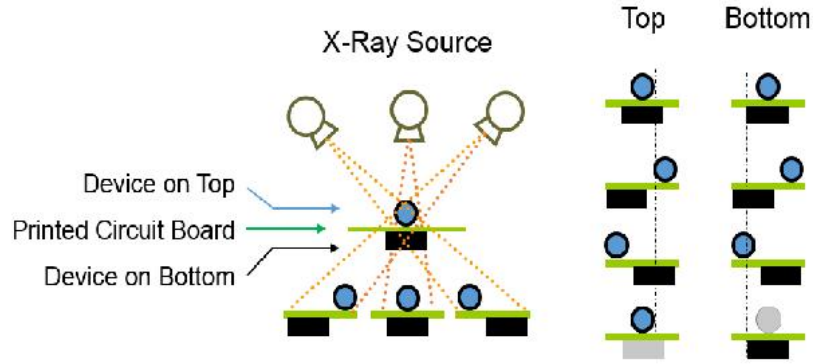
- The X-ray mean is rotating. Deflection coils in the tube cause the photon stream to pass through the panel at a rotated angle.
- Agilent 5DX uses X-ray Laminography.

Example - 5DX Laminography



- Laminography required oblique angle images for 3D reconstruction.
- Relative motion of the source, detector, and object under inspection generates a cross-sectional image by smearing into the background, all information not located on a predetermined plane-of-focus (POF).
- Because only 1 areal mode camera is used, multiple angle images can only being obtained by moving both x-ray spot and detector.

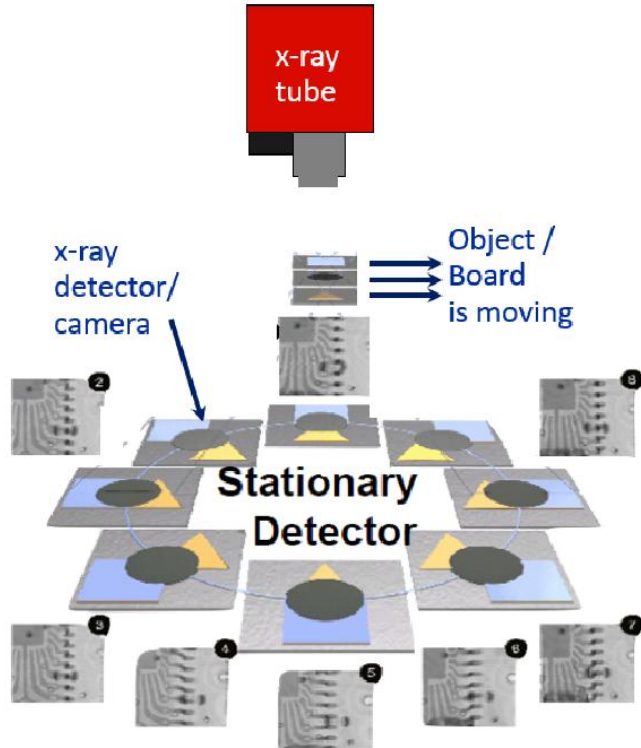
Digital Tomosynthesis (V810)



Digital Tomosynthesis X-ray Image Reconstruction:

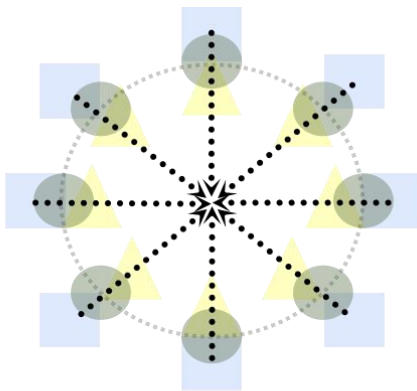
- Uses Transmission X-ray, and capture the images in different angle.
- The images from different angle is then combined to reconstruct 3D images.
- ViTrox V810 AXI system uses Digital Tomosynthesis X-ray Image Reconstruction Technology.

Example - Tomosynthesis

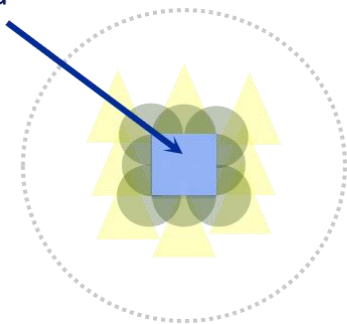


- As opposed to 5DX's Laminography, object / board is moving in Tomosynthesis during image acquisition while x-ray spot and detector (camera) are not.
- To obtain different angle images, multiple TDI cameras are installed at different angle with respect to the object

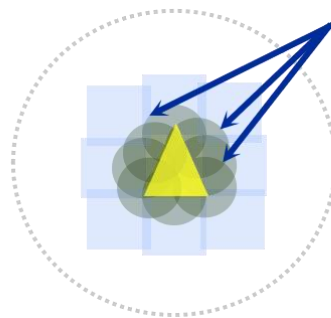
Image Reconstruction Process



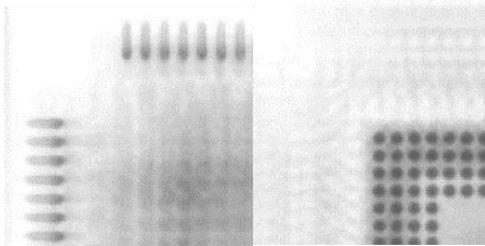
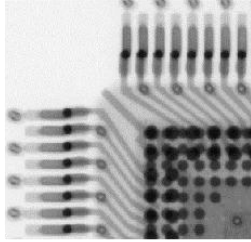
Reconstructed
blue square



Artifacts



Comparing 2D and 3D images



Comparing 2D and 3D images:

- Image on the top is 2D image
- Image on the bottom are 3D image
- All images are from the same PCB, same region.

Notice:

- 2D image contains information from the top and bottom of the panel
- 3D image contains information from only one side of the panel
- Edges in 2D image are crisp
- Edges in 3D image are fuzzy
- If there are voids on the solder ball
 - 3D image can detect but 2D image don't

Digital Reconstruction

- Uses industry standard techniques for oblique image reconstruction.
- Allows for arbitrary number of in focus planes from a single acquisition sequence.
- V810 system uses multiple shift and combine methodologies.
 - Different methods for defect image optimization: shift+average, shift+ordered statistics, etc
 - Multiple standard statistical noise reduction techniques, such as, averaging, double correlated sampling, etc.

Lesson 2

V810 Model and Configuration

V810 Machine

Machine Type	Standard (Std)				XXL		S2EX		XLT S2		XLL		XLTM		S3
Series	S1		S2		S1	S2	-		S2		-		1	2	-
Magnification (um)	19	19 & 11	19	19 & 11	19 & 13		23 / 19 & 11		19 / 15 & 11 OR 19 / 10 & 7.5		19 / 15 & 11 OR 19 / 10 & 7.5		19 / 15 & 11		22 / 19 15 / 12 10* / 7*
PSP	Non	1	2	2.1	1	2	2.1	2	2.1	2	2.1	2	2.1	2.1	2.1
Support Pin	-				Optional		Optional		-		-		-	-	-

Remark:

- **Red** = Obsolete
- **Green** = Optional
- **Yellow** = Programming option, 23+11um, 19+11um or 19+11um (S2 XLT only)
- **Blue** = Basic package

*Std and XXL assembled with pneumatic VM position change VM and S2 XLT and Mini running with motorized.

V810 Machine



Standard
(Std S2)



XXL S2



S2EX



XLT / XLTM

V810 Machine



XLL



V810 XLW



V810 S3

V810i Machine Software version Matrix

		2.0	2.2	3.0	3.1	3.2	4.0	4.1	5.0	5.2	5.3	5.4	5.5	5.6	5.6c	5.7	5.8	5.9	5.10	5.11	5.12	5.13	5.14	5.15	5.16	5.17	5.18	5.18.2	5.18.3	5.19
Hardware Feature	AKD																			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Biometric Login																		✓						✓	✓	✓	✓	✓	✓
	Camera Gen3																		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Camera Gen3.1																			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Camera Gen3.2																										✓	✓	✓	✓
	Camera Gen3.3																											✓	✓	✓
	Cast Iron Beam PIP																							✓	✓	✓	✓	✓	✓	✓
	Curtain Sensor																													
	QMP																								✓	✓	✓	✓	✓	✓
	Support Pin																										✓	✓	✓	✓
	S200																										✓	✓	✓	✓
	Variable Magnification								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	ViCAT IO																										✓	✓	✓	✓
	Panasonic Motion Drive																												✓	✓

V810 Machine (S2EX)



System footprint	1566mmx2145mmx1972mm
Max PCB size	609.6 mm x 482.6 mm (24"x19")
Min PCB size	76.2 mm x 76.2 mm (3" x 3")
PCB thickness	0.5 - 7mm
Max PCB weight	4.5kg
Top clearance (Variable Mag)	50 mm @ 23 μ m resolution 38 mm @ 19 μ m resolution 11 mm @ 11 μ m resolution 38 mm @ 10.5 μ m resolution# 11 mm @ 6 μ m resolution#
Bottom Clearance	70 mm
Weight	3800kg
Server OS	Windows 10 Enterprise LTSC
Features	PSP, Optical slow Mechanical stop

V810 Machine (XLT/ XLTM)



	XLT	XLTM
System footprint	2240mmx2460mmx1980mm	2048mmx2460mmx1980mm
Max PCB size	965.2mmx660.4mm (38"x26")	865.0mmx660.4mm (34"X 26")
Min PCB size	76.2mmx76.2mm (3"x3")	
PCB thickness	0.5 - 12.7mm	
Max PCB weight	15kg	
Top clearance (Variable Mag)	50 mm @ 19 µm resolution 31 mm @ 15 µm resolution 13 mm @ 11 µm resolution 31 mm @ 10 µm resolution# 13 mm @ 7.5 µm resolution#	
Bottom Clearance	80mm	
Weight	~6500kgs	~6000kgs
Server OS	Windows 10 Enterprise LTSC	
Features	PSP, Optical slow Mechanical stop	PSP, Optical slow Mechanical stop, enhanced rail with enhanced belt

V810 Machine (S3)

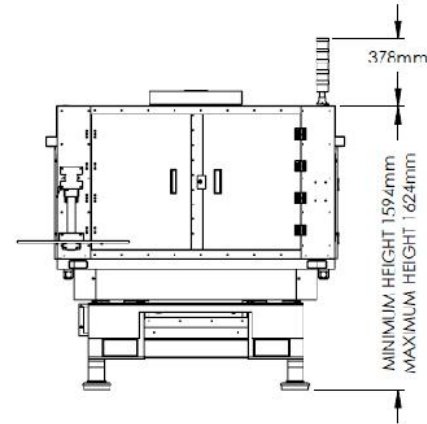
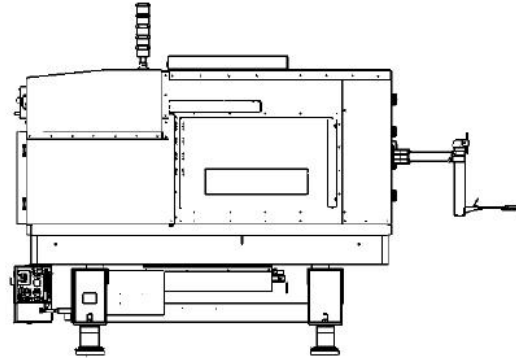
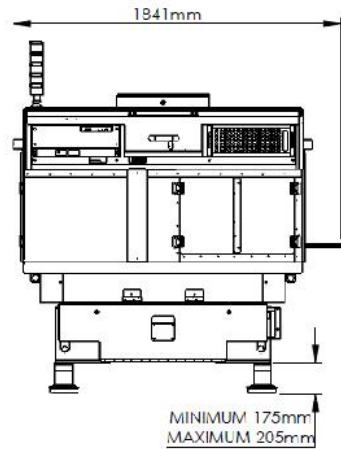
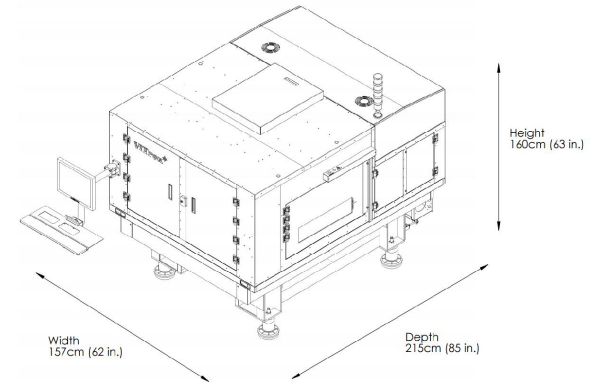
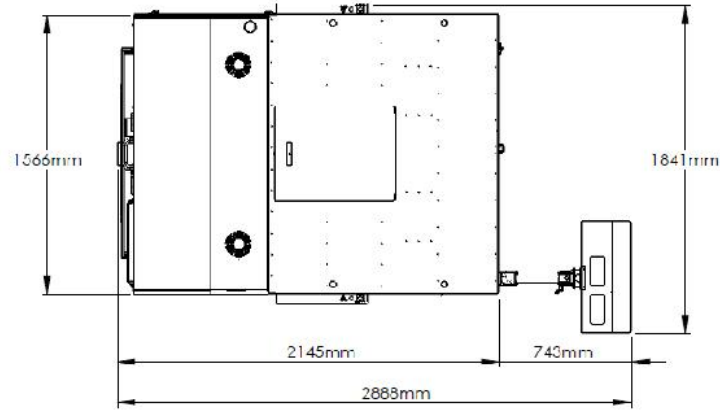


System footprint	1835mmx2185mmx2162mm
Max PCB size	725mmx482.6mm (28.5"x19")
Min PCB size	63.5mmx63.5mm (2.5"x 2.5")
PCB thickness	0.5 - 7mm
Max PCB weight	4.5kg
Top clearance (Variable Mag)	50 mm @ 22 μ m resolution 42 mm @ 19 μ m resolution 26 mm @ 15 μ m resolution 11 mm @ 12 μ m resolution 26 mm @ 10 μ m resolution# 11 mm @ 7 μ m resolution#
Bottom Clearance	80mm
Weight	~4800kgs
Server OS	Windows 10 Enterprise LTSC
Features	PSP, Optical slow Mechanical stop enhanced rail with enhanced belt Bridging Mechanism

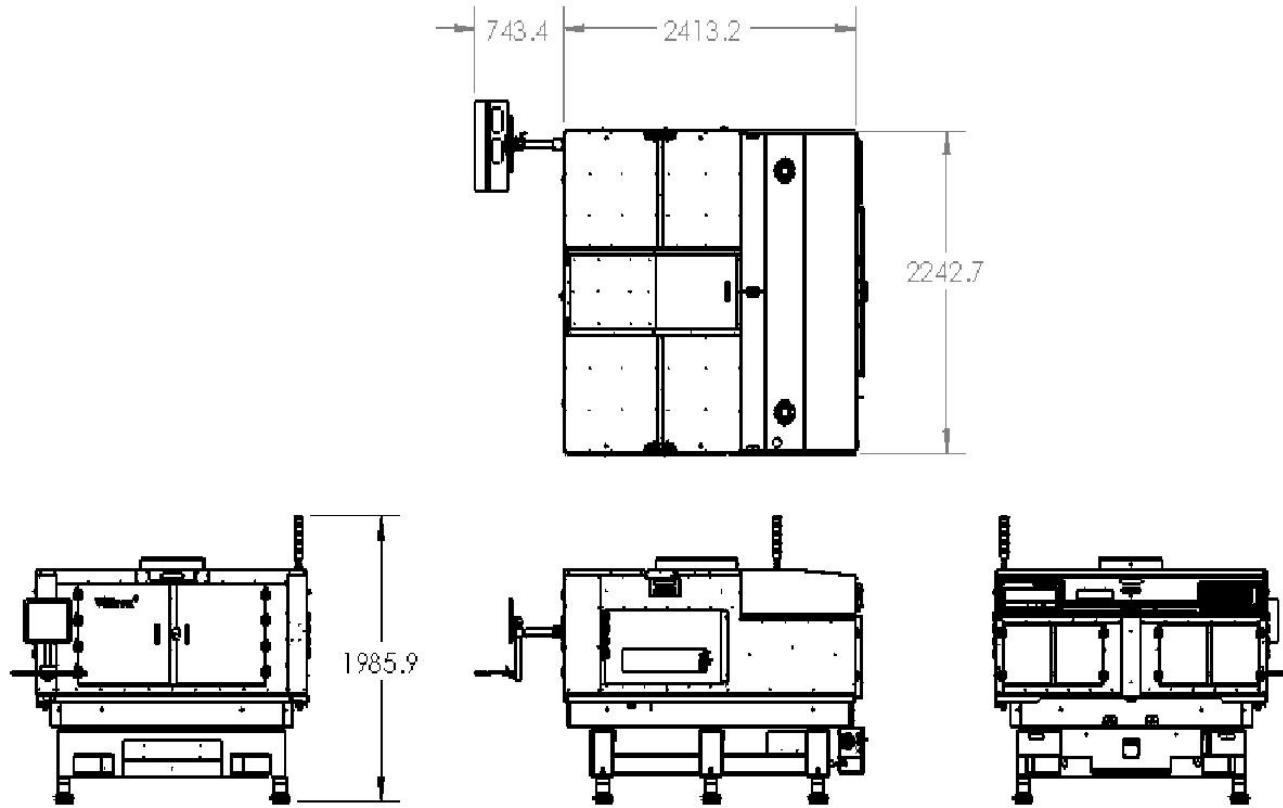
Lesson 3

Machine Overview

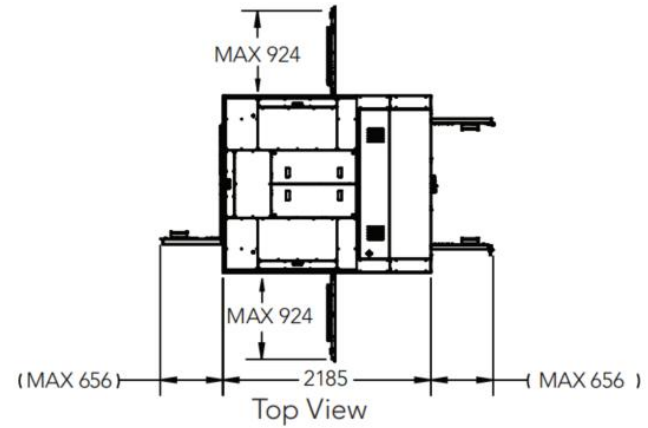
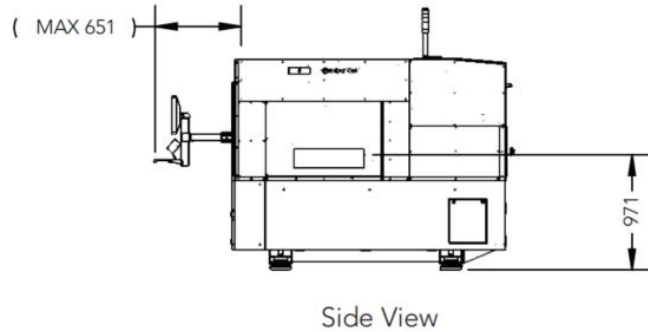
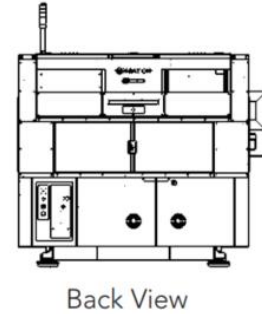
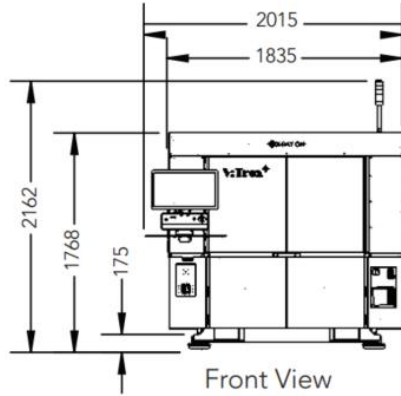
Machine Dimension - S2EX



Machine Dimension - XLT



Machine Dimension - S3

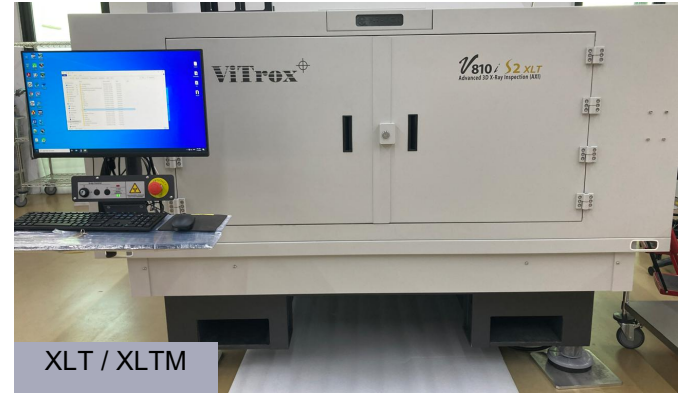


Lesson 3.B

Machine Outer View

Front View

No	Description
1	Cabinet
2	Monitor / Monitor Arm
3	Operator Control Panel
4	Fork Truck Holes
5	Front Door
6	PDU (S3)



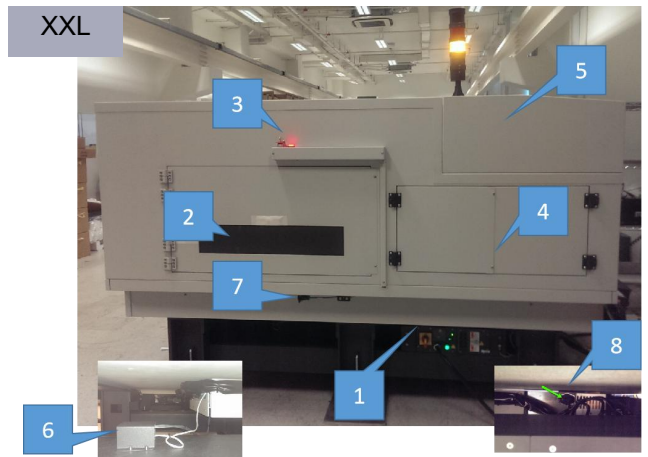
Left View

No	Description
1	Controller Rack
2	Air pressure regulator
3	Left Outer Barrier
4	Emergency Off Switch (EMO Button)
5	External SMEMA Connection (Upstream Conveyor)



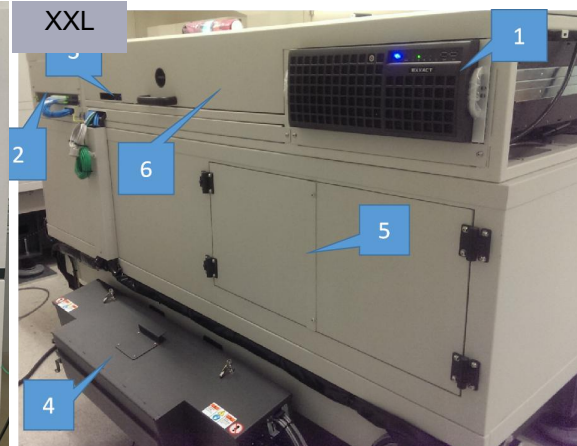
Right View

No	Description
1	Power Distribution Unit (PDU)
2	Right Outer Barrier
3	Emergency Off Switch (EMO Button)
4	Right Door (L/R)
5	Server Controller Rack (Open/Closed)
6	External SMEMA Connection (Downstream)
7	Camera Trigger Board Switch (On/Off)
8	Break out Board (BOB)



No	Description
1	Super Micro Server
2	Atom PC(RMS server)
3	Gigabyte Switch
4	System Status & Control Assembly (SSCA)
5	Rear Door (L/R)
6	Server Rack Top cover handle & lock
7	Internet/Intranet LAN Cable
8	Bar-code COM Cable (R2S32)

Rear View



Inner Rear View

No	Description
1	POE Switch
2	Atom PC
3	Center IRP Fan
4	Slow Speed Controller
5	Gas Dampers
6	H-tube 24V DC Power Supply
7	Z Axis Stepper Motor Power Supply
8	Filter Height Sensor Module Controller
9	Filter Height Module Controller Power Supply



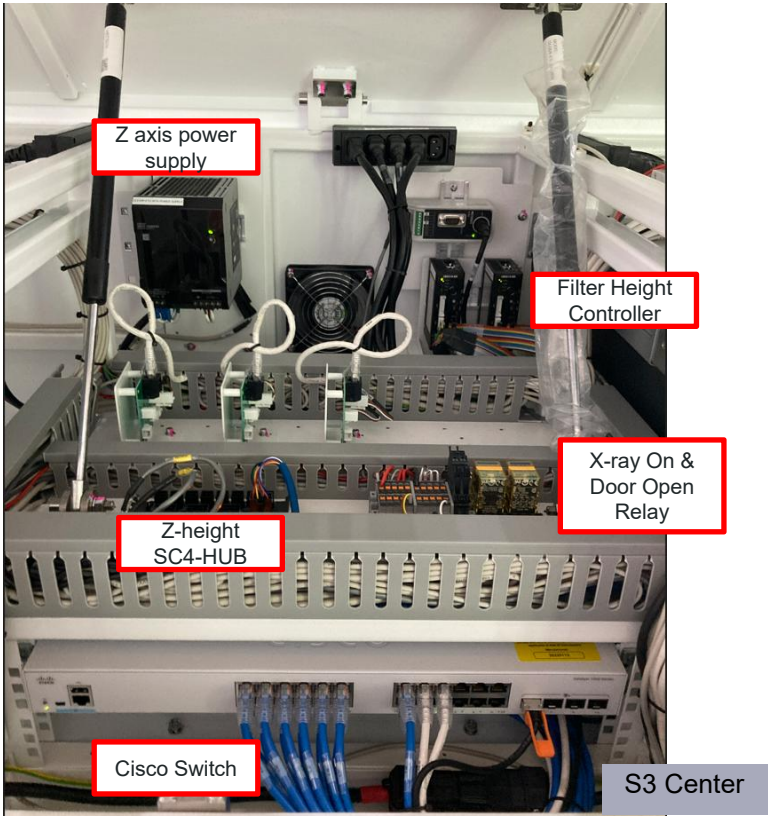
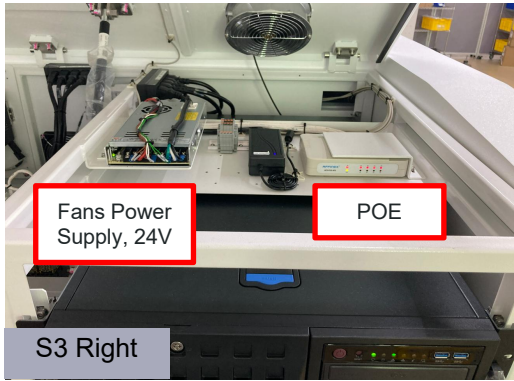
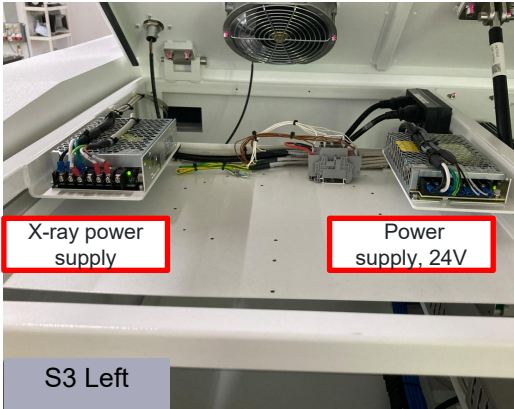
S2EX



XLT

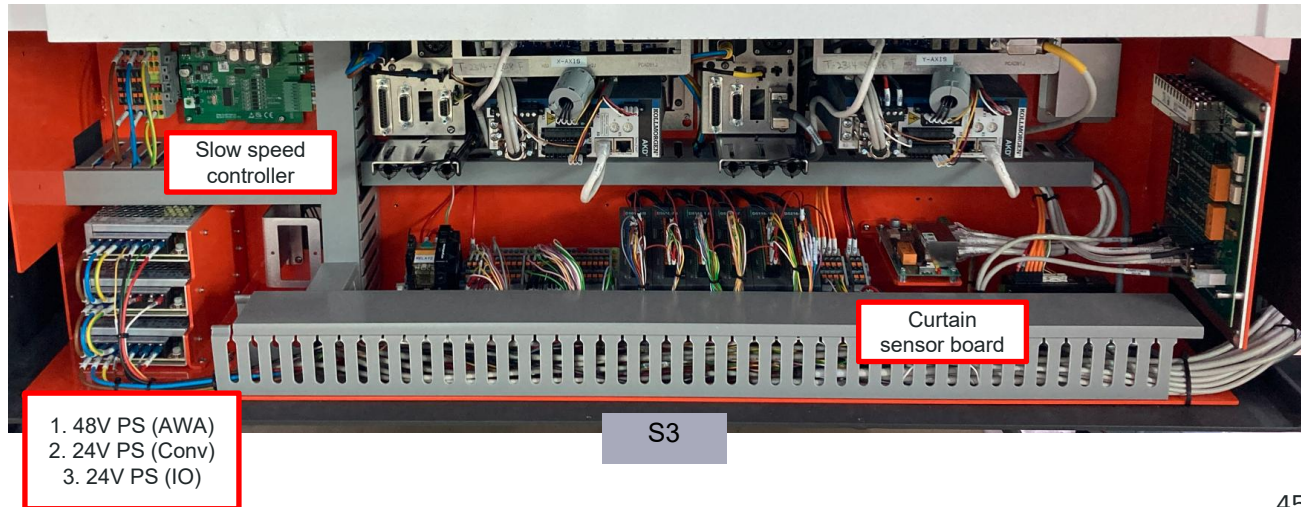


Inner Rear View (S3)



SSCA View

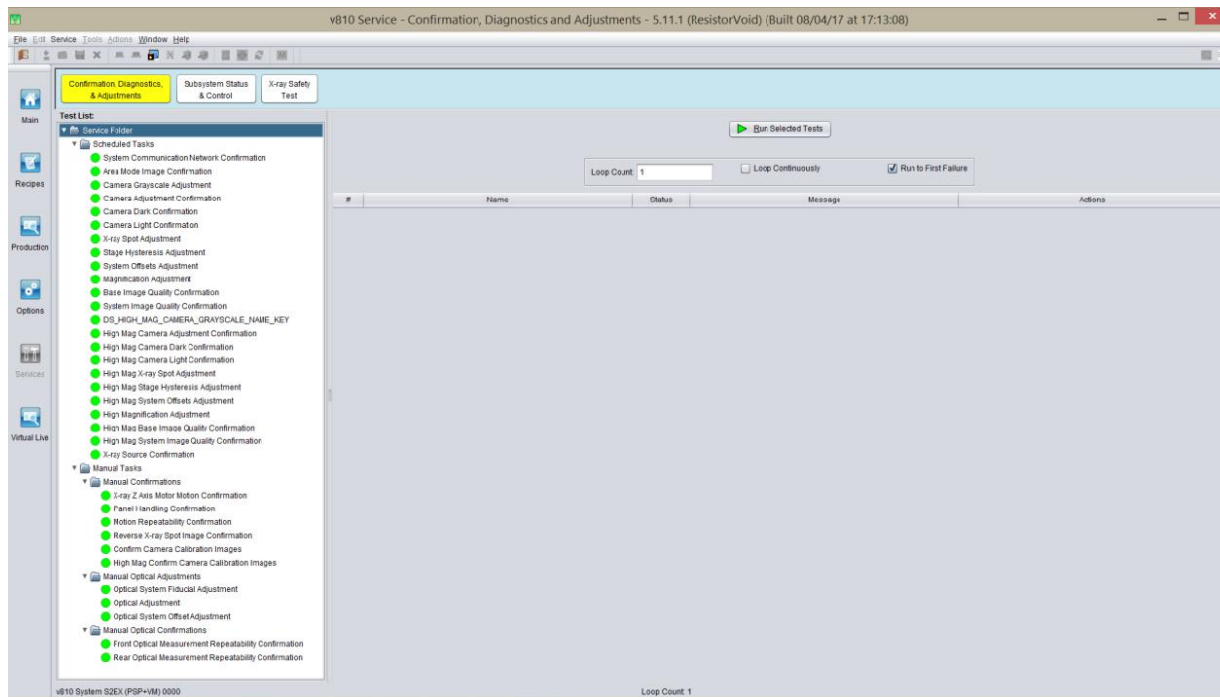
No	Description
1	X axis driver
2	Y axis driver
3	Vicat IO
4	AWA moons driver
5	Conveyor Motor On Relay
6	Conveyor Motor direction relay
7	Interlock board



Lesson 4.B

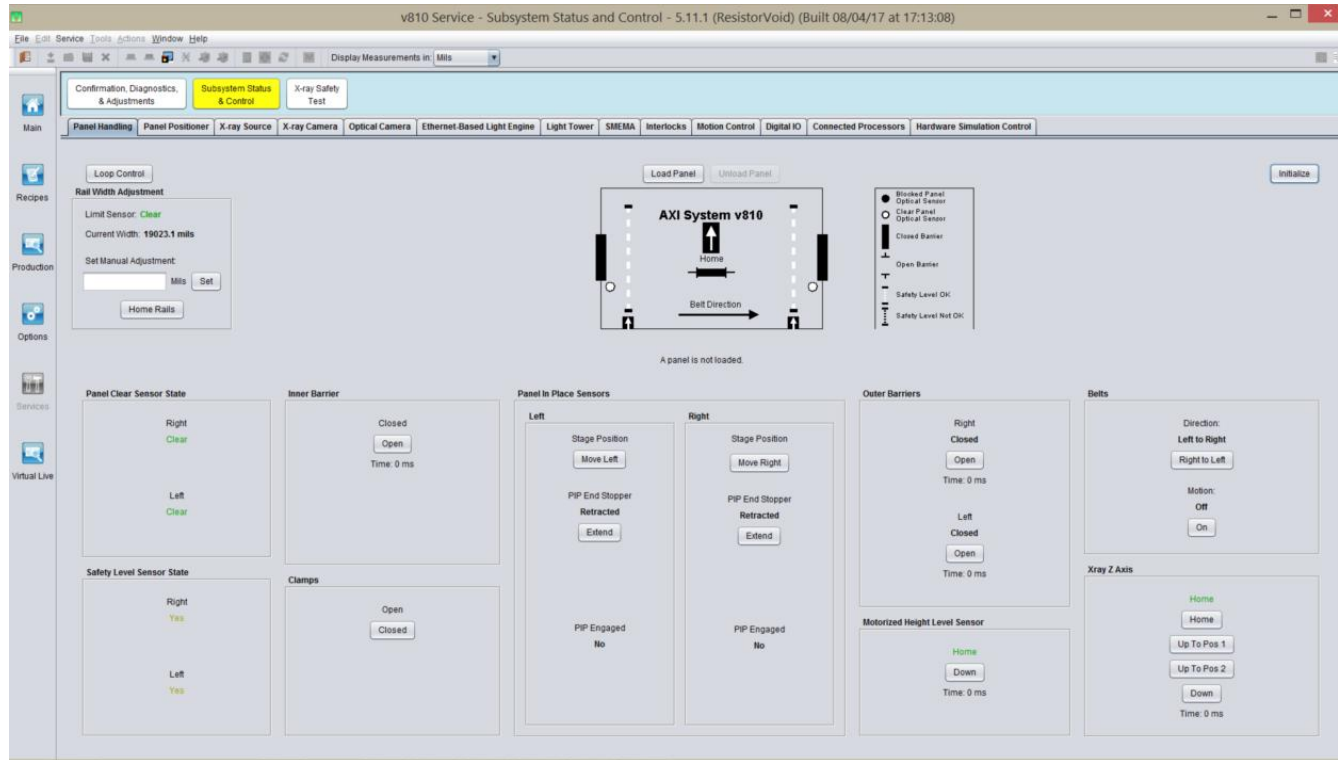
Service Tab

System GUI - Service Tab - Confirmation Diagnostic and Adjustment Tab



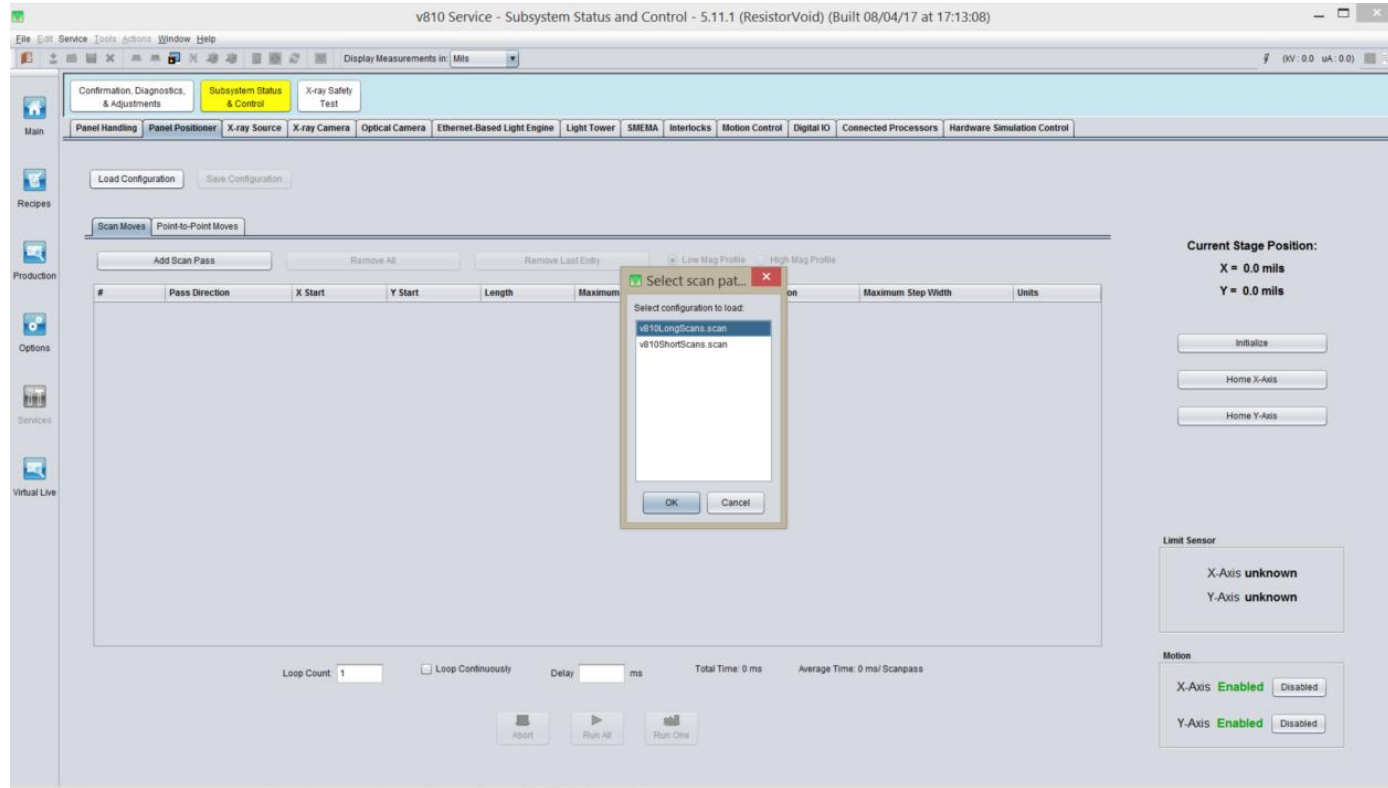
- Confirmation and adjustment before perform board inspection.
- An automatic periodic checking will be carried out at an interval time to make sure the system condition is within intended specification. And if found not within intended specification, the software will adjust its calculation to make sure the inspection performance is consistent with previous inspections. Until a point that calculation adjustment is impossible, the machine will prompt message stating the machine needs to be serviced.

System GUI - Service Tab - Subsystem Status & Control Page - Panel Handling Tab



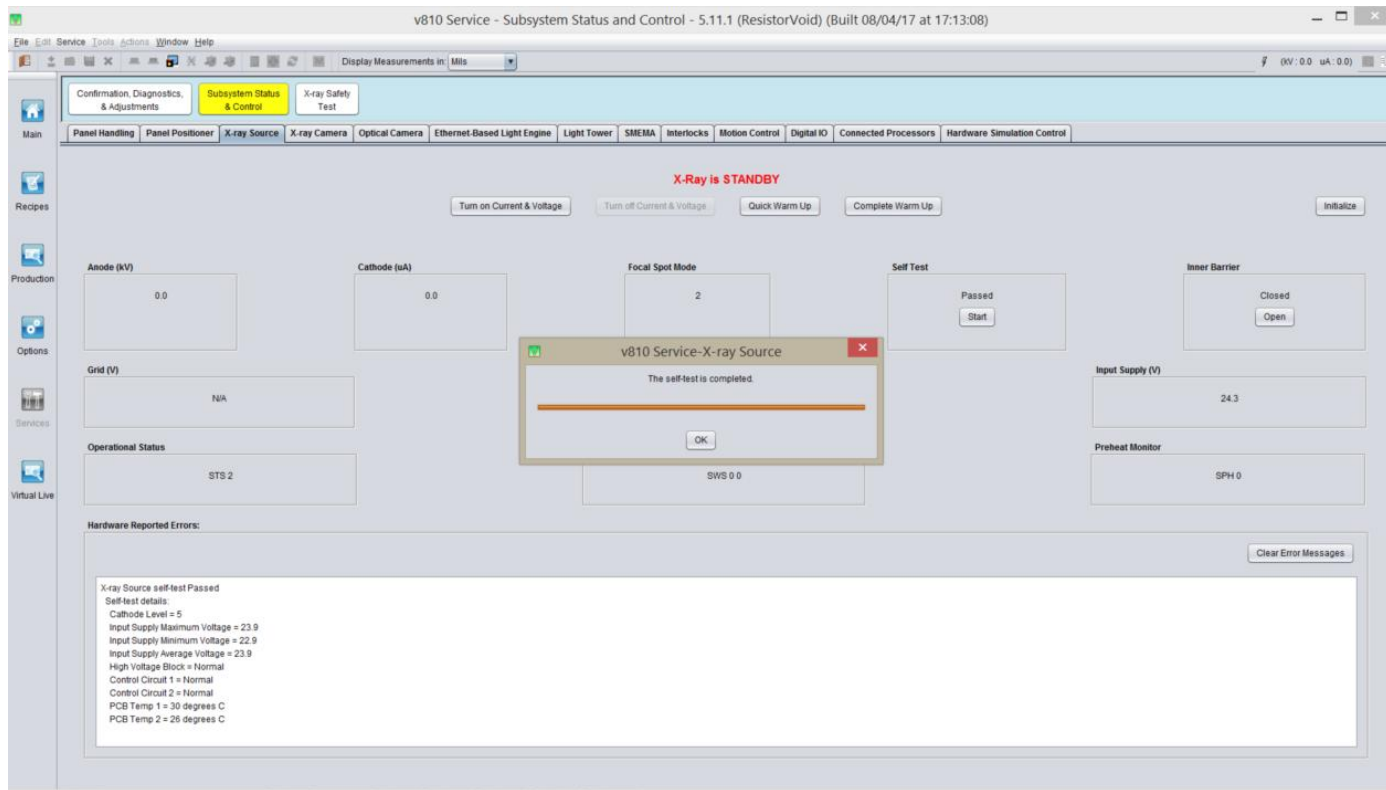
- Unable to initialize if Panel Clear Sensor input value incorrect or interlock break.
- Same function as DIO tab with simply view (#manually toggle output bit and feedback the input value)
- Show movement speed for inner barrier/outer barrier/dual mag
- Home AWA and manually set width as per wish
- Manual load panel with set width

System GUI - Service Tab - Subsystem Status & Control Page - Panel Positional Tab



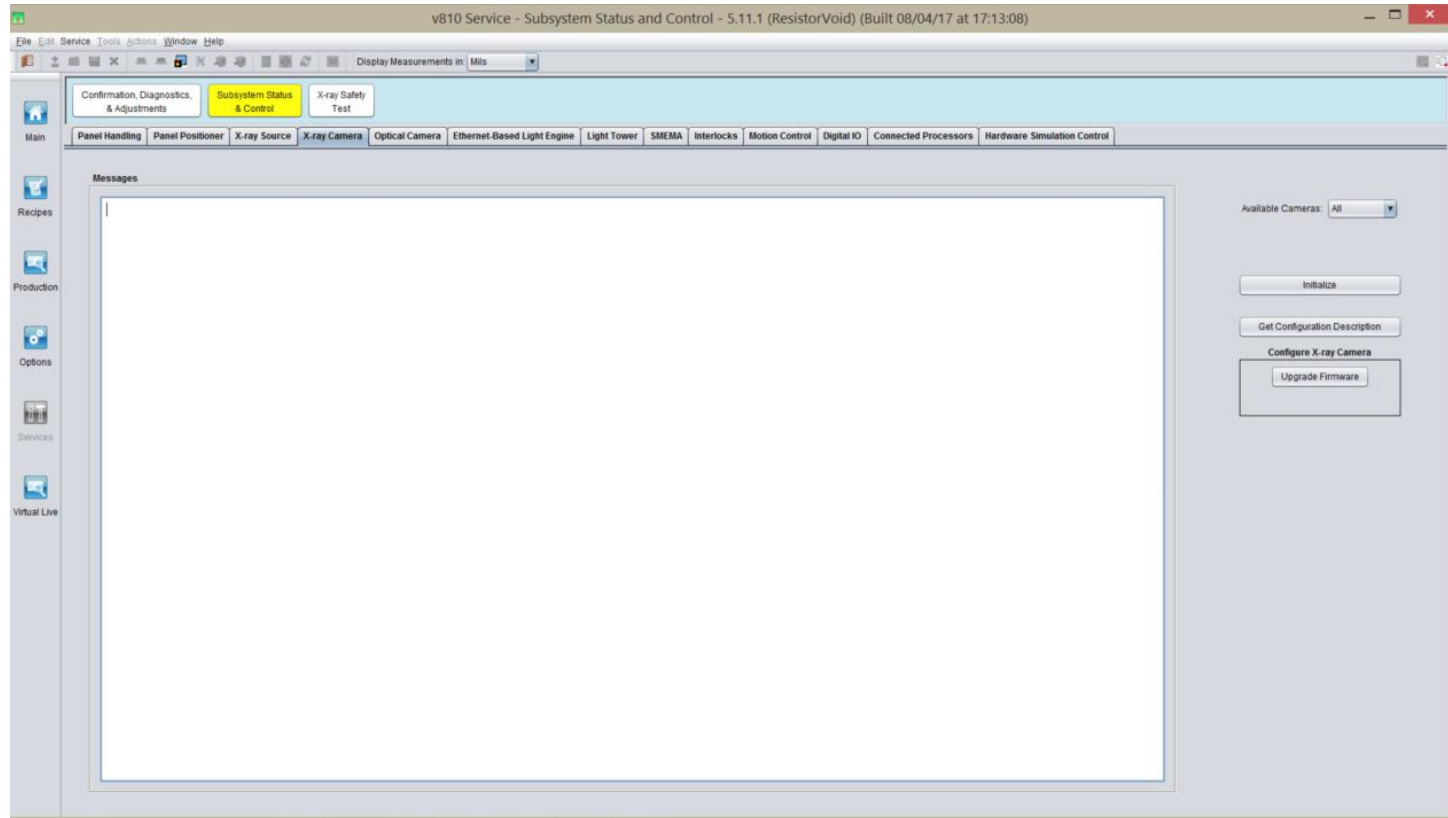
- Unable initialize if Interlock break or motion hardware failure
- Motion repeat-ability checking
- Current stage position information
- Allow individual home/ move stage position with given command

System GUI - Service Tab - Subsystem Status & Control Page - XRay Source Tab



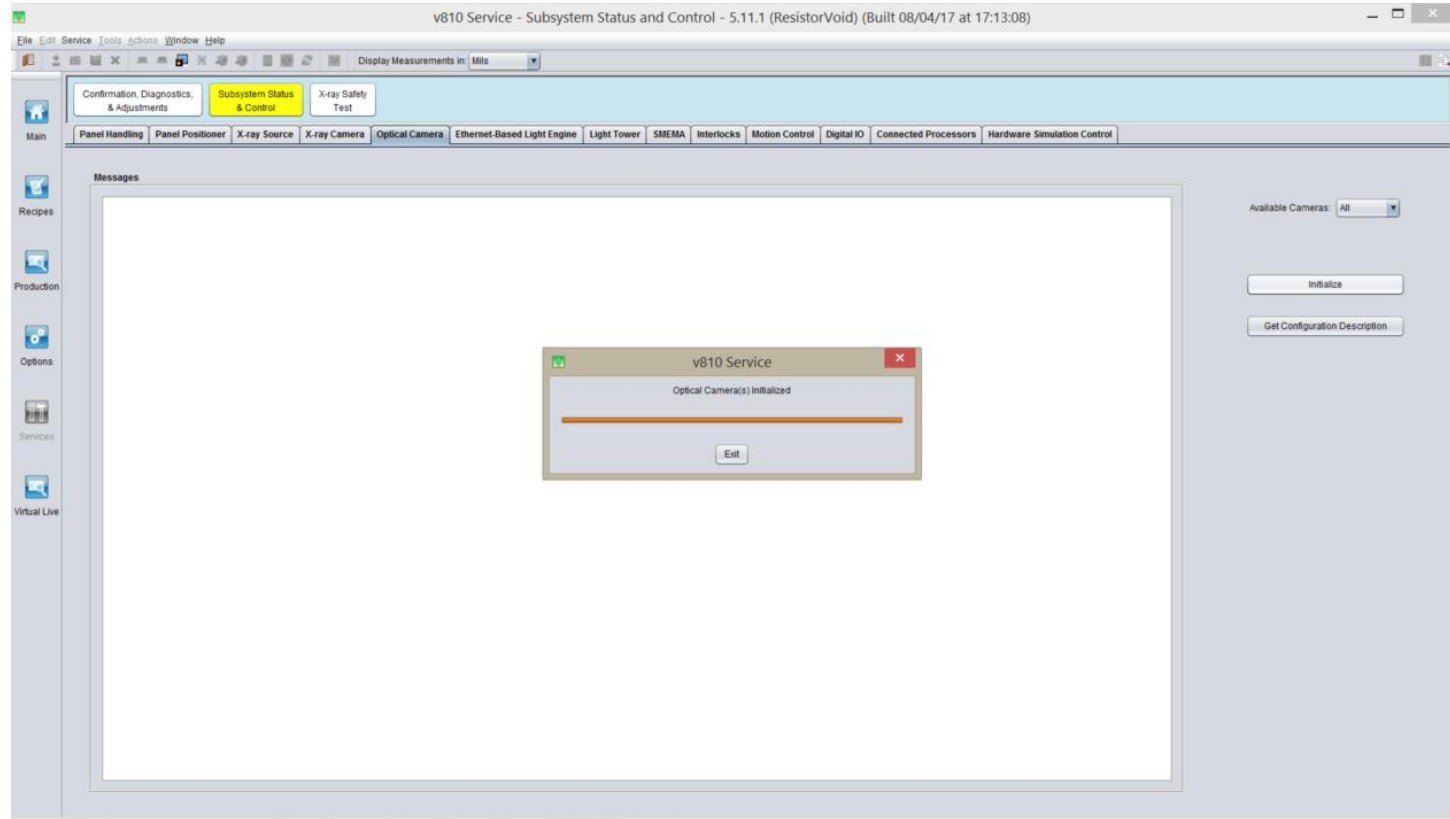
- On/Off tube power
- Perform tube warm up
- Perform tube self test and provide tube current status
- Unable initialize if interlock break or tube not in ready status.

System GUI - Service Tab - Subsystem Status & Control Page - XRay Camera Tab



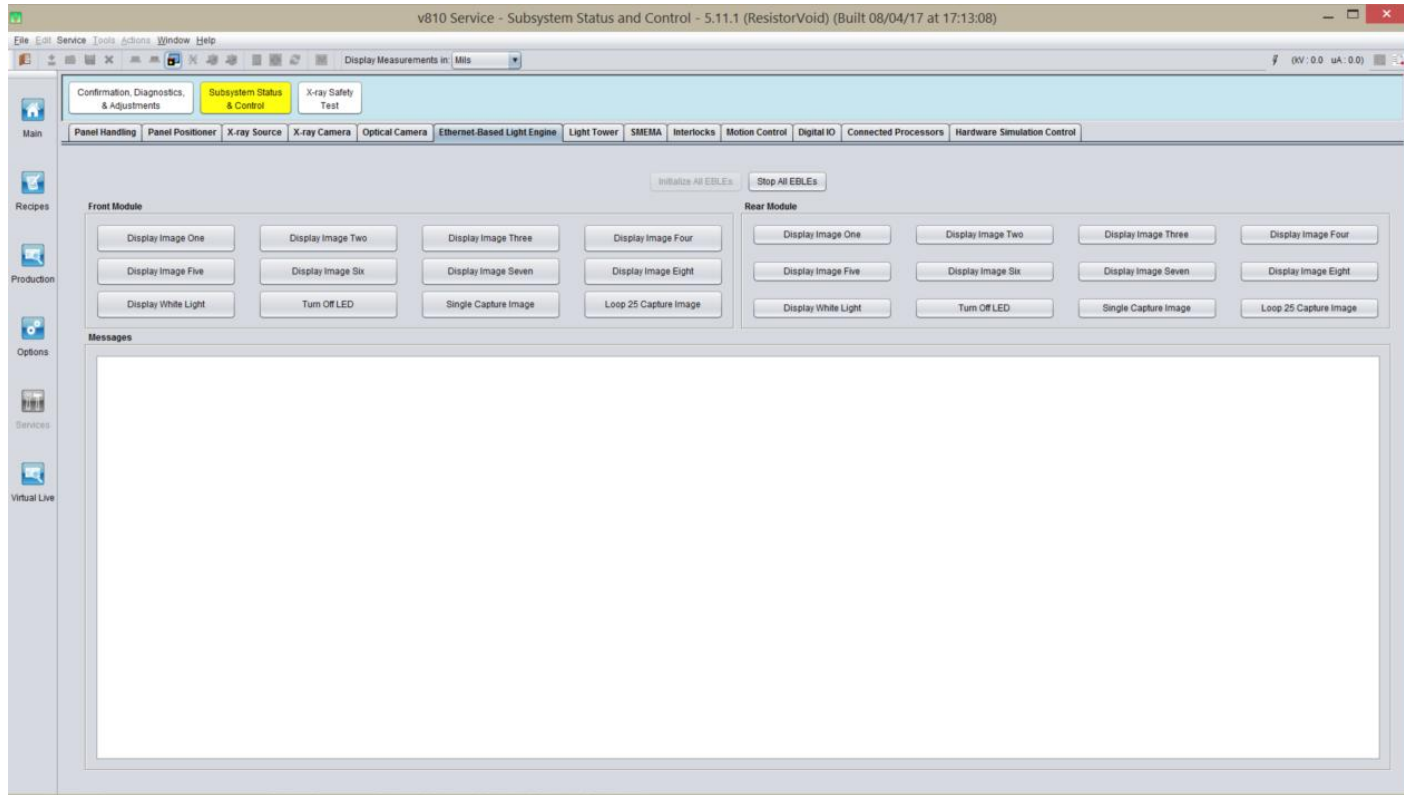
- Camera firmware upgrade base on camera version
 - # If change or swap AXI X-Ray camera, the Camera upgrade firmware MUST be performed
- Check camera communication status with system (all cameras or individual)
- Unable Initialize or perform firmware upgrade if connection error

System GUI - Service Tab - Subsystem Status & Control Page - Optical Camera Tab



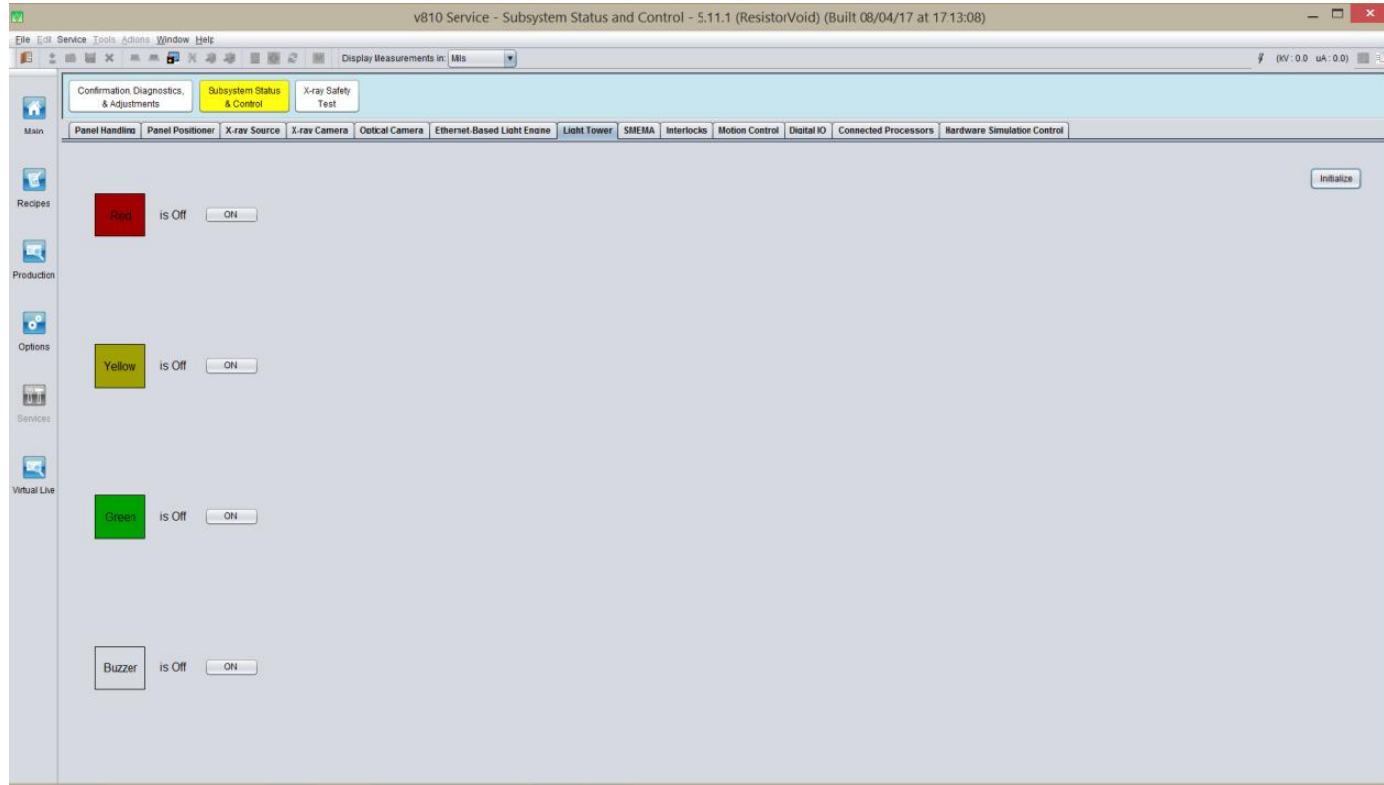
- Communication status verification for PSP optical camera with main server
- Unable initialize if hard ware failure.
(# Possible Root cause; USB cable not seat firmly or optical camera power lost)

System GUI - Service Tab - Subsystem Status & Control Page - EBLE Tab



- PSP 2 projector main tab
- Manually trigger project with selected fringe pattern for focusing verification
- Unable initialize if power lost at projector or trigger cable connection issue

System GUI - Service Tab - Subsystem Status & Control Page - Light Tower Tab



- Manual trigger Tower light functional for verification purpose

System GUI - Service Tab - Subsystem Status & Control Page - SMEMA Tab

The screenshot shows the 'v810 Service - Subsystem Status and Control' application window. The title bar indicates the version is 5.11.1 (ResistorVoid) and it was built on 08/04/17 at 17:13:08. The interface includes a menu bar (File, Edit, Service, Tools, Actions, Window, Help) and a toolbar with icons for various functions. A sidebar on the left contains icons for Main, Recipes, Production, Options, Services, and Virtual Live. The main content area has a top navigation bar with tabs: Confirmation Diagnostics & Adjustments, Subsystem Status & Control (selected), X-ray Safety Test, Panel Handling, Panel Positioner, X-ray Source, X-ray Camera, Optical Camera, Ethernet Based Light Engine, Light Tower, SMEMA, Interlocks, Motion Control, Digital IO, Connected Processors, and Hardware Simulation Control. Below the navigation bar, the 'SMEMA' section displays a status table and a process flow diagram. The status table lists four conditions: TRO (System Ready to Accept, is False), TRI (Downstream System Ready to Accept, is False), SRO (Panel From System Available, is False), and SRI (Panel From Upstream System Available, is False). Each condition has a 'True' button next to it. The process flow diagram shows the interaction between an 'Upstream Machine', the 'v810' unit, and a 'Downstream Machine'. Arrows indicate the flow of signals: 'TRO Ready To Accept' and 'SRI Board Available' from the Upstream Machine to the v810; 'TRI Ready To Accept' and 'SRO Board Available' from the v810 to the Downstream Machine; and 'GPO' and 'UPO' from the v810 to the Downstream Machine. Below the diagram, the 'Test Results' section includes three radio buttons: 'Set GPO to Passed', 'Set GPO to Failed', and 'Set UPO to Untested'. A table displays the test results for GPO and UPO.

Test Result	GPO	UPO
Untested	False	True
Passed	True	False
Failed	False	False

➤ SMEMA connection verification

System GUI - Service Tab - Subsystem Status & Control Page - Interlocks Tab

v810 Service - Subsystem Status and Control - 5.11.1 (ResistorVoid) (Built 08/04/17 at 17:13:08)

File Edit Service Tools Actions Window Help

Display Measurements in: Mils SERVICE MODE IS NOT ENABLED

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control X-ray Safety Test

Main Panel Handling Panel Positioner X-ray Source X-ray Camera Optical Camera Ethernet-Based Light Engine Light Tower SMEHA Interlocks Motion Control Digital IO Connected Processors Hardware Simulation Control

Recipes Production Options Services Virtual Live

Status	Name
?	(1) Access Panel #A, Chain 1 (J2)
?	(2) Access Panel #A, Chain 2 (J4)
?	(3) Access Panel #B, Chain 1 (J3)
?	(4) Access Panel #B, Chain 2 (J5)
?	(5) Camera Enclosure #J, Chain 1 (J2)
?	(7) Camera Enclosure #J, Chain 2 (J4)
?	(9) Access Panel #C, Chain 1 (J2)
?	(10) Access Panel #C, Chain 2 (J4)
?	(11) Access Panel #D, Chain 1 (J3)
?	(12) Access Panel #D, Chain 2 (J5)
?	(13) Left Outer Barrier Presence, Chain 1 (J3)
?	(14) Left Outer Barrier #H, Chain 1 (J2)
?	(15) Left Outer Barrier Presence, Chain 2 (J5)
?	(16) Left Outer Barrier #H, Chain 2 (J4)
?	(17) Right Outer Barrier #I, Chain 1 (J2)
?	(18) Right Outer Barrier Presence, Chain 1 (J3)
?	(19) Right Outer Barrier #I, Chain 2 (J4)
?	(20) Right Outer Barrier Presence, Chain 2 (J5)
?	(21) X-Ray Tube Enclosure #G, Chain 1 (J2)
?	(23) X-Ray Tube Enclosure #G, Chain 2 (J4)
?	(25) Access Panel #E, Chain 1 (J2)
?	(26) Access Panel #E, Chain 2 (J4)
?	(27) Access Panel #F, Chain 1 (J3)
?	(28) Access Panel #F, Chain 2 (J5)

Legend

✓ Closed ↑ Open ? Status Unknown

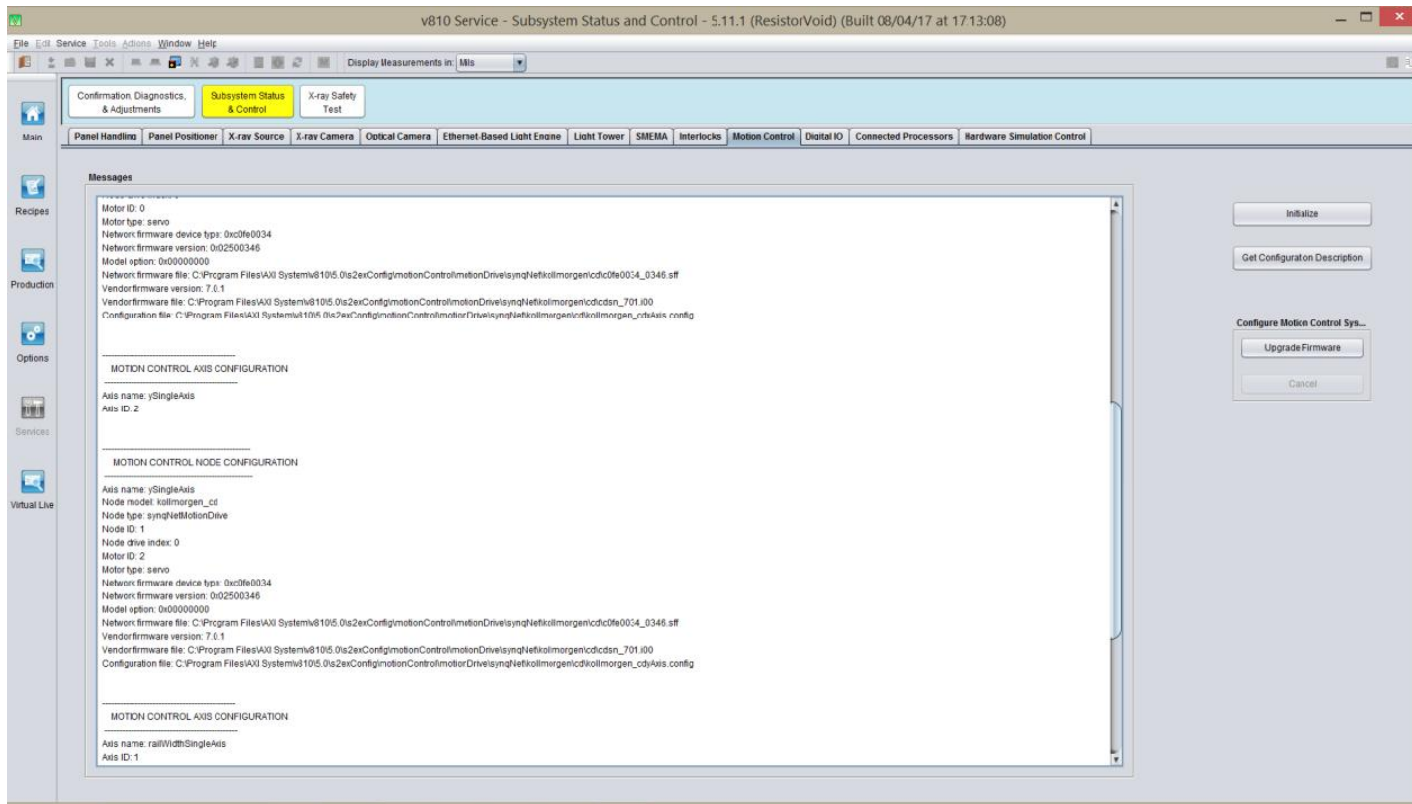
Top View

Interlock Junction Board Chain 1 Chain 2 Hardware Revis...

Power is OK Closed Closed 6

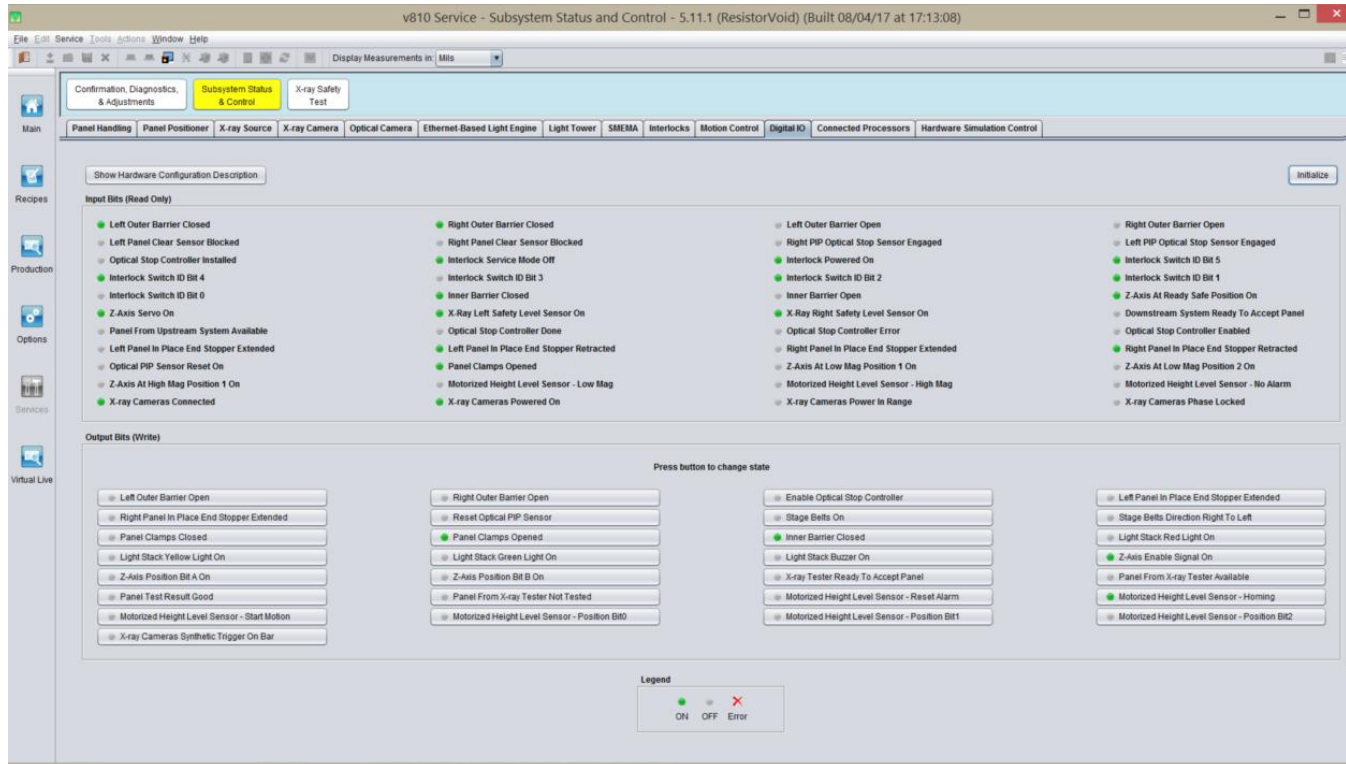
- System Interlock checking interface, when enable service mode, will trigger interlock break affected area
- Unable to initialize once DIO tab initialized. Direct switch to Service Mode will shown.

System GUI - Service Tab - Subsystem Status & Control Page - Motion Control Tab



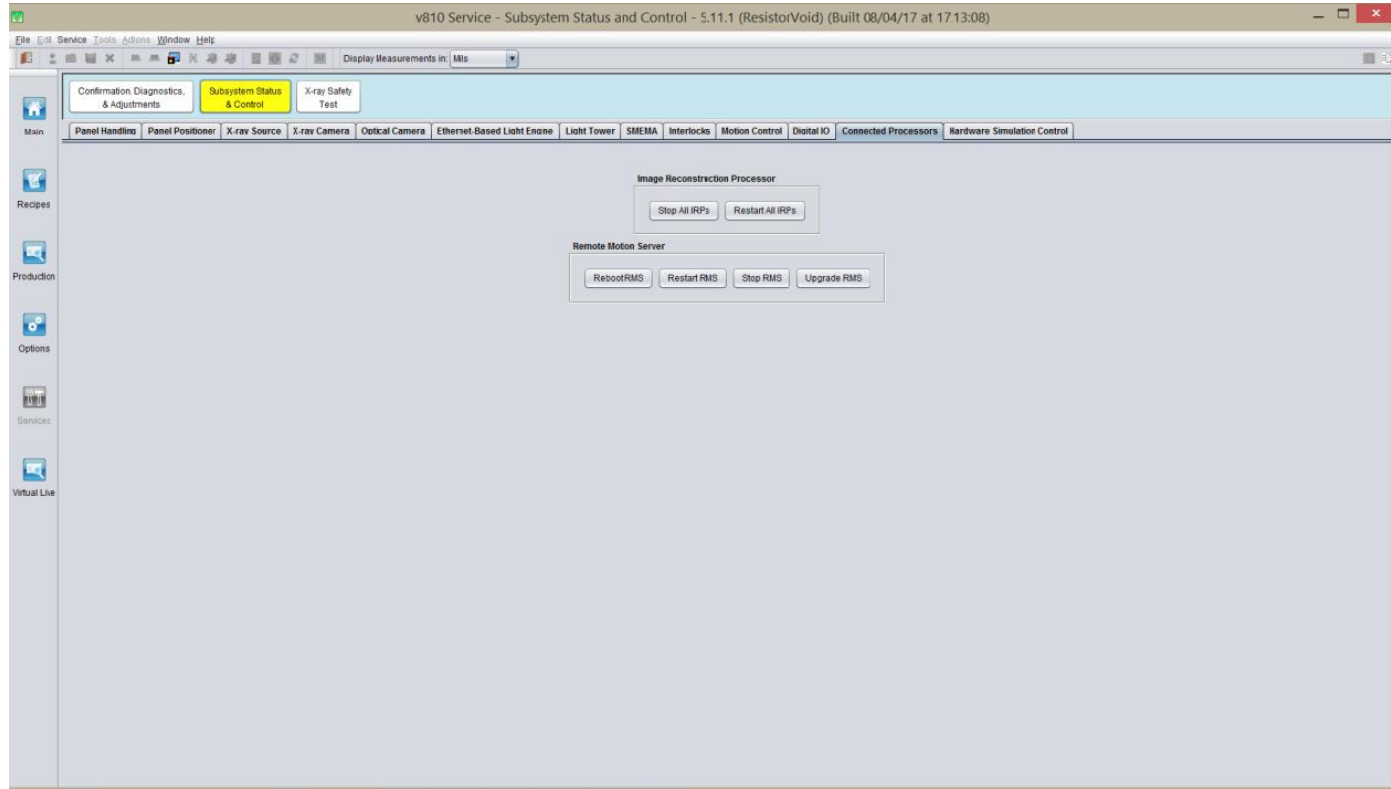
- Check the connection for X and Y axis Kollmorgen kits with main server
- Perform Kollmorgen kits firmware upgrade (download motion profile to kollmorgen) each time change Kollmorgen kit or servo motor

System GUI - Service Tab - Subsystem Status & Control Page - Digital I/O Tab



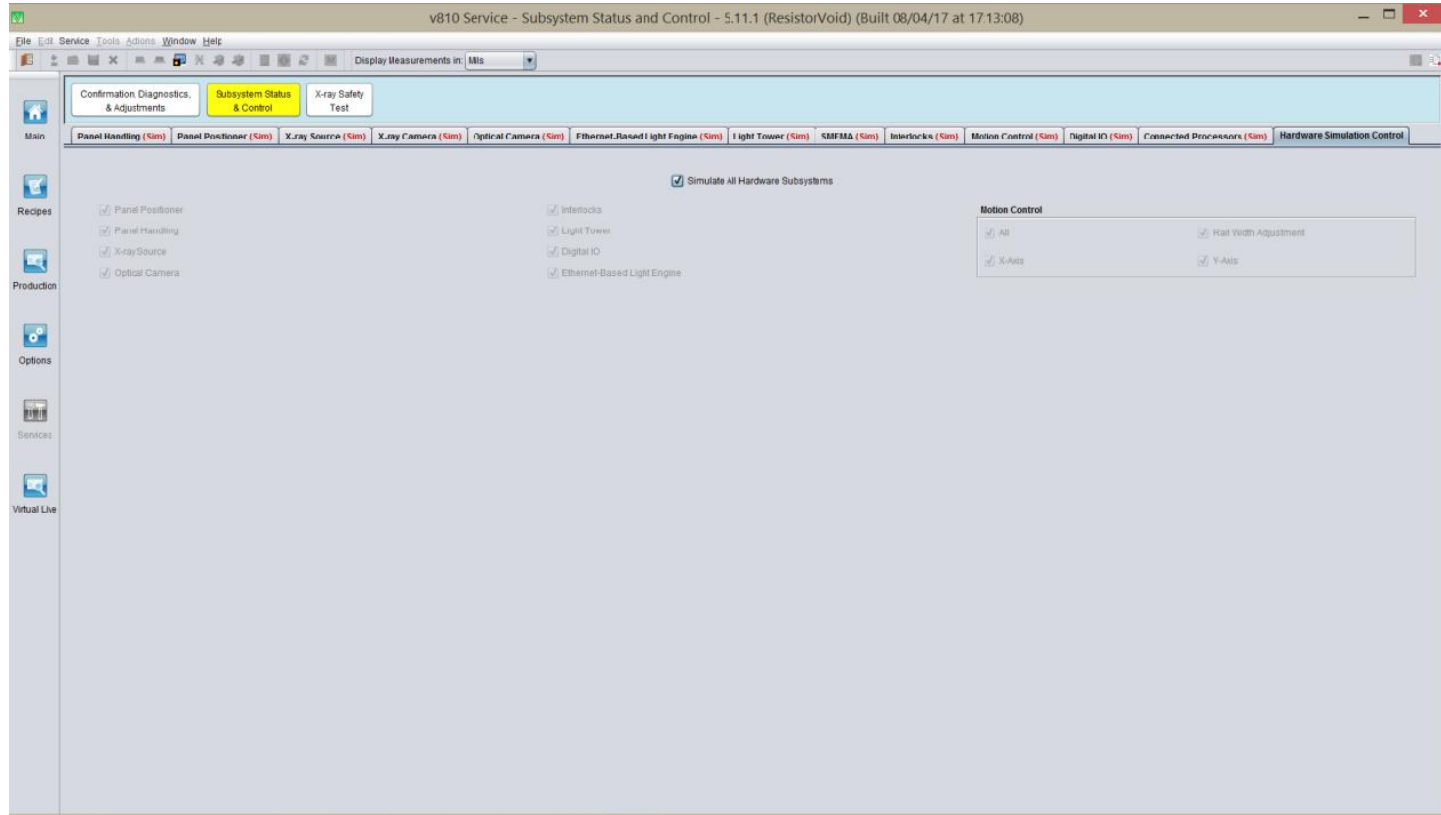
- Direct control signal for acting cylinder and sensor input reading without checking interlock signal
- Unable initialize if Atom PC or SSCA didn't initialize or do not power ON the device

System GUI - Service Tab - Subsystem Status & Control Page - Connected Tab



- Main controller tab to Stop/Restart and update RMS and IRP software
- Check communication between IRP, RMS(Atom PC) and main server connectivity
- Unable restart id devices didn't power On.

System GUI - Service Tab - Subsystem Status & Control Page - Hardware Simulation Tab



- Simulation hardware condition at Offline Learning Station (OLP)

System GUI - Service Tab - XRay Safety Test Tab

v810 Service - X-ray Safety Test - 5.11.1 (ResistorVoid) (Built 08/04/17 at 17:13:08)

File Edit Service Tools Actions Window Help

Confirmation Diagnostics, & Adjustments Subsystem Status & Control **X-ray Safety Test**

X-ray Source Voltage (kV) Current (µA)

Panel Positioner X-Axis (mils) Y-Axis (mils)

Rail Width Width (mils)

Magnification Low Mag

Inner Barrier Closed

Outer Barriers Left Closed Right Closed

ViTrox v810 X-ray Safety Test Procedure

Please refer to the ViTrox v810 X-ray Safety Test Procedure document for detailed safety, regulatory and operation instructions.

Purpose

This procedure provides instructions to check the operation of the safety interlocks and to measure the X-ray emission at all points outside the external surfaces of the v810 AX system cabinet.

Training Requirements

Attending the ViTrox v810 CE Training Class is required before performing the v810 X-ray Safety Test.

Procedure and Report Usage

According to the X-ray Safety Test Procedure, an X-ray Safety Test Report must always be completed. Be sure you have the most recent copy of the ViTrox v810 X-ray Safety Test Procedure and X-ray Safety Test Report.

Pass Specification

Readings must not exceed 0.05 mR/hr (0.5 µSv/hr) when measured with the ion chamber. Any confirmed reading above 0.05 mR/hr (0.5 µSv/hr) shall be considered a failure of the cabinet X-ray system.

Required Tools

Note: Review this ViTrox v810 X-ray Safety Test Procedure document for the most up to date tool requirements.

ViTrox v810 X-ray Safety Test Report
A Geiger-Mueller (GM) meter
An ion chamber

Click Next to select a safety test.

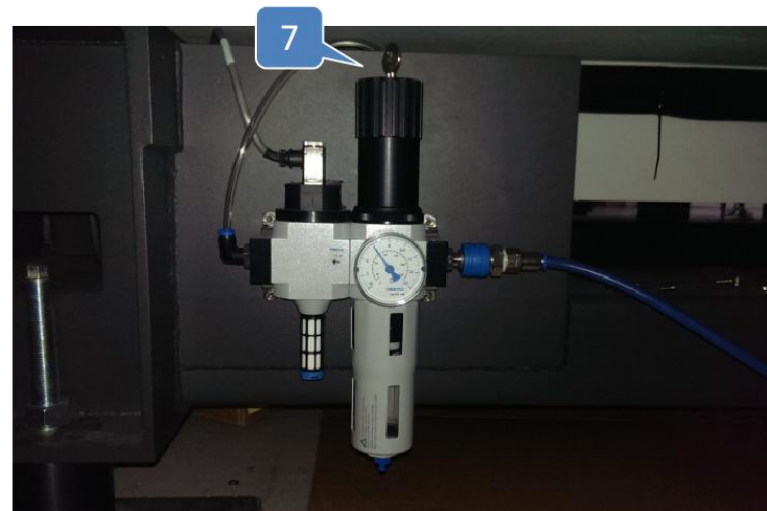
Cancel Back Next Finish

- Main tab to perform X-Ray Survey with manual select test method, Low Power test and/or High Power Test

Lesson 5

Machine Overview

System Startup - PDU Power and Air Regulator Enable



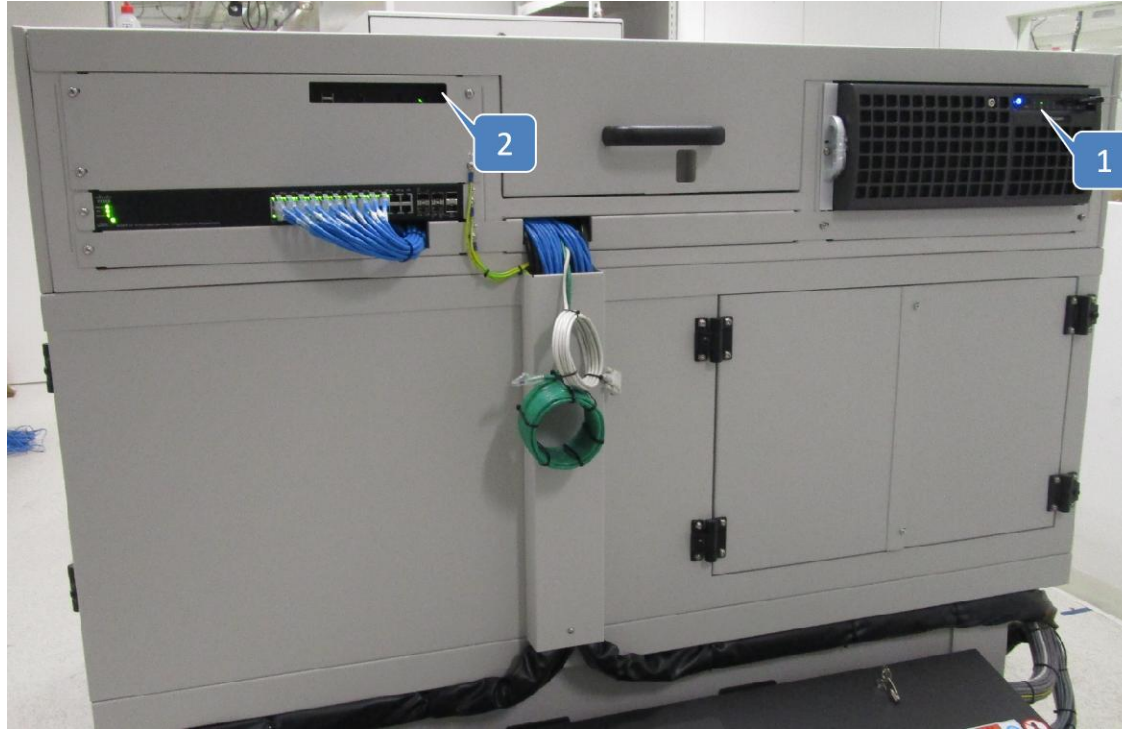
1. Release EMO (if enable) and ensure all access door closed
2. Power on incoming power at plant facilities
3. Switch main power inlet switch to “On” status
4. Turn On Unswitch port power on switch
5. Press “Power On” Button
6. Switch On Camera Controller Assemble Power
7. Set Air Inlet to 80 PSI (Lock key to turn knob and done, release lock)

System Startup - SSCA/MCA Box Power Enable



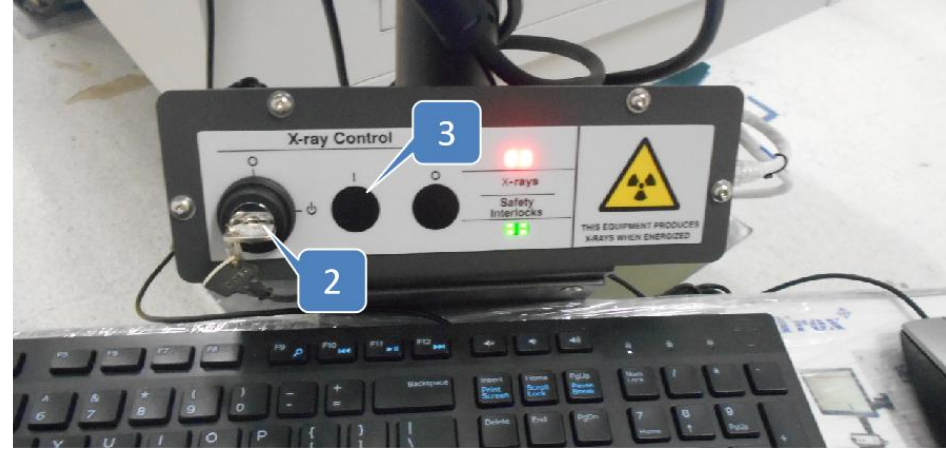
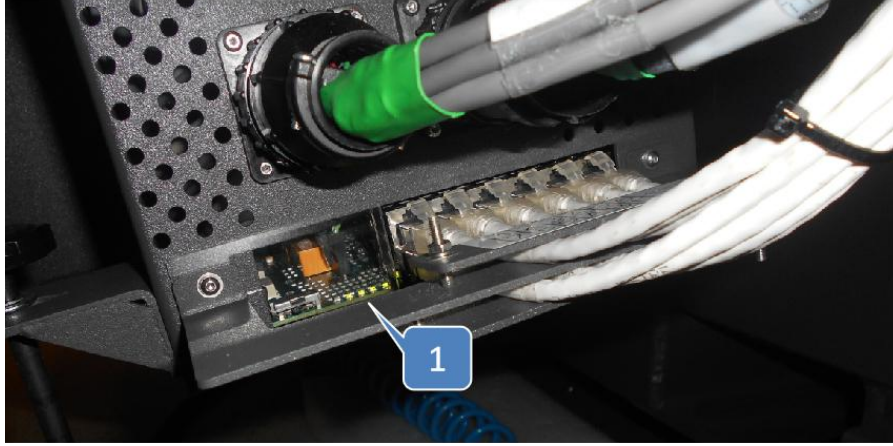
1. Switch SSCA/MCA Box main power inlet to “On” position

System Startup - Controller(PC) Power On



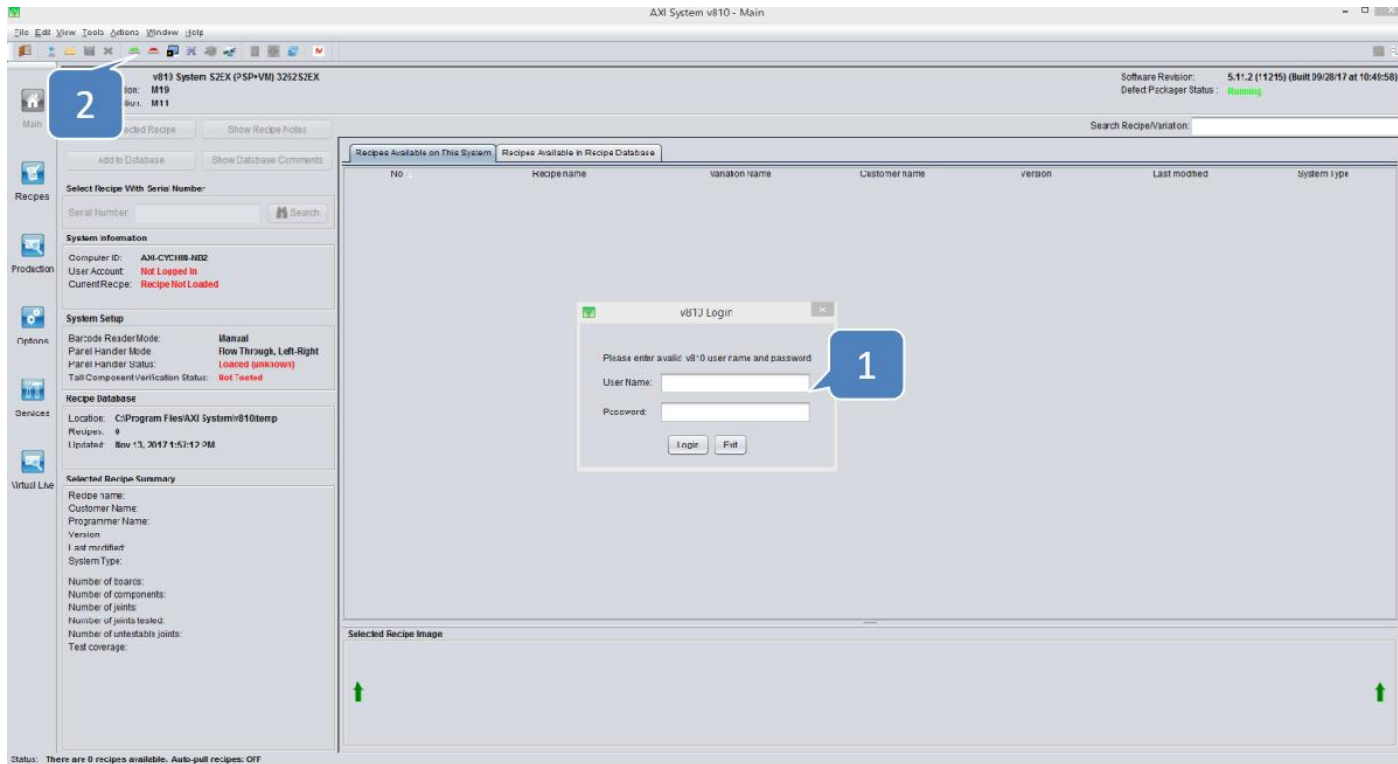
1. Turn on main controller
2. Turn on RMS (Atom PC)

System Startup - Operator Console Panel Active



1. Visual check at Interlock card, 4 green LED should emitter once machine power on if no any interlock break.
2. Twish the key to "On" position.
3. Trigger the "X-Ray On" button and Red LED will emitter

System Startup - GUI Login and System Initialization



1. Launch V810 GUI and log in with desire user account and password
2. Click the start-up button or shortcut key "Ctrl+U" and wait for initialization complete

System Startup - Xray Tube Initialization Period

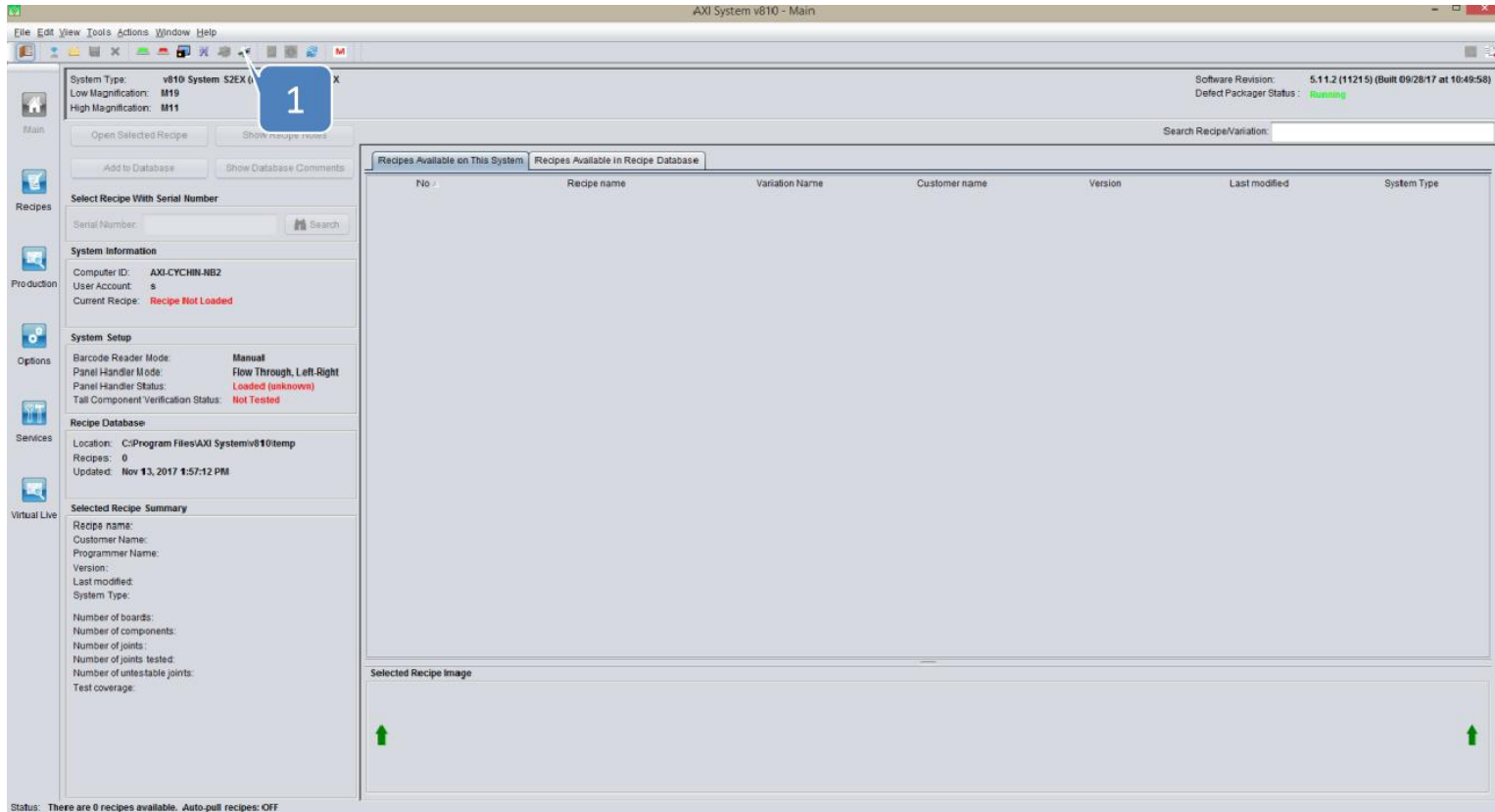
Step	Tube voltage (kV)	Tube current (μ A)	Warm-up time (minutes)		
			Pattern 1	Pattern 2	Pattern 3
			8 hours to 1 month after power off	1 to 3 months after power off	3 months or more after power off
1	27	0	1	5	10
2	54	30	1	5	30
3	81	90	3	6	20
4	108	150	3	7	30
5	121	220	3	7	20
6	130	300	4	10	10
Total (minutes)			15	40	120

- During machine start-up X-ray tube will self perform warm-up operation
- Warm-up operation pattern varies according to the period X-ray tube had been off
- X-ray tube do have controller board itself and shielded
- Warm-up operation will self define by the X-ray tube

Lesson 5.B

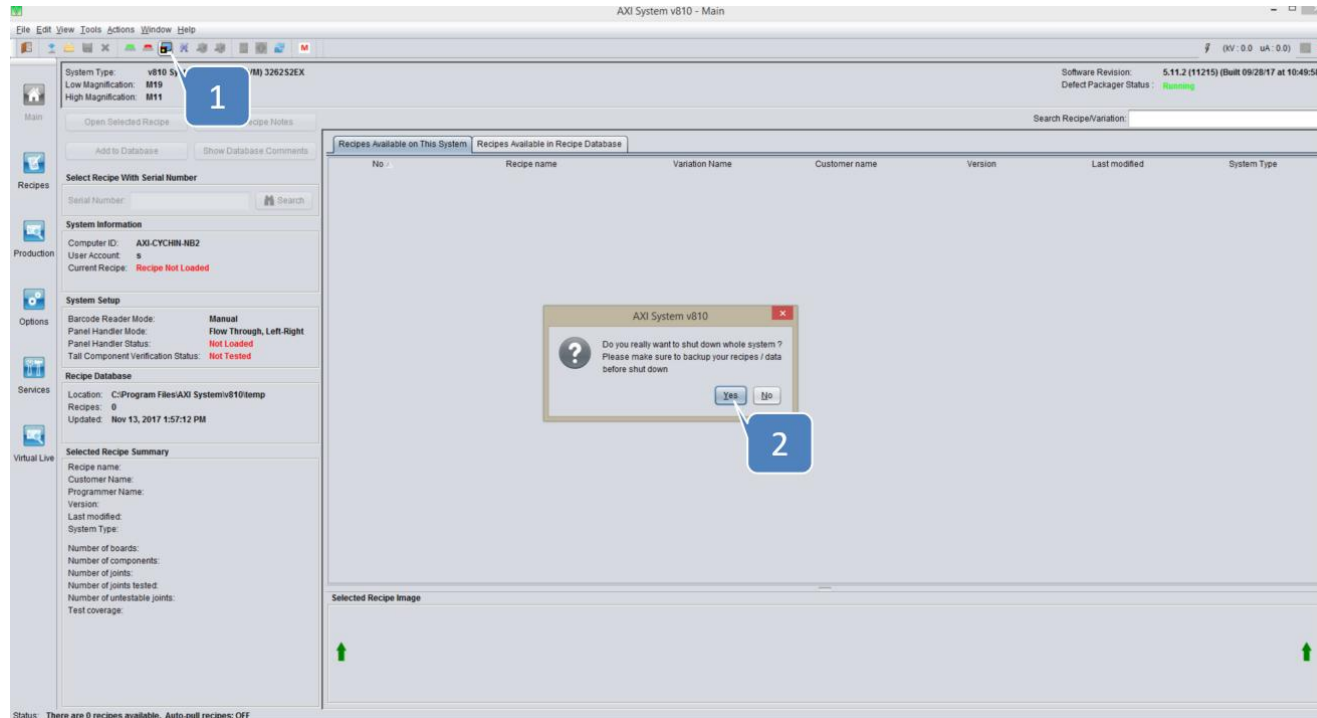
System Shutdown

System ShutDown - Unload Panel



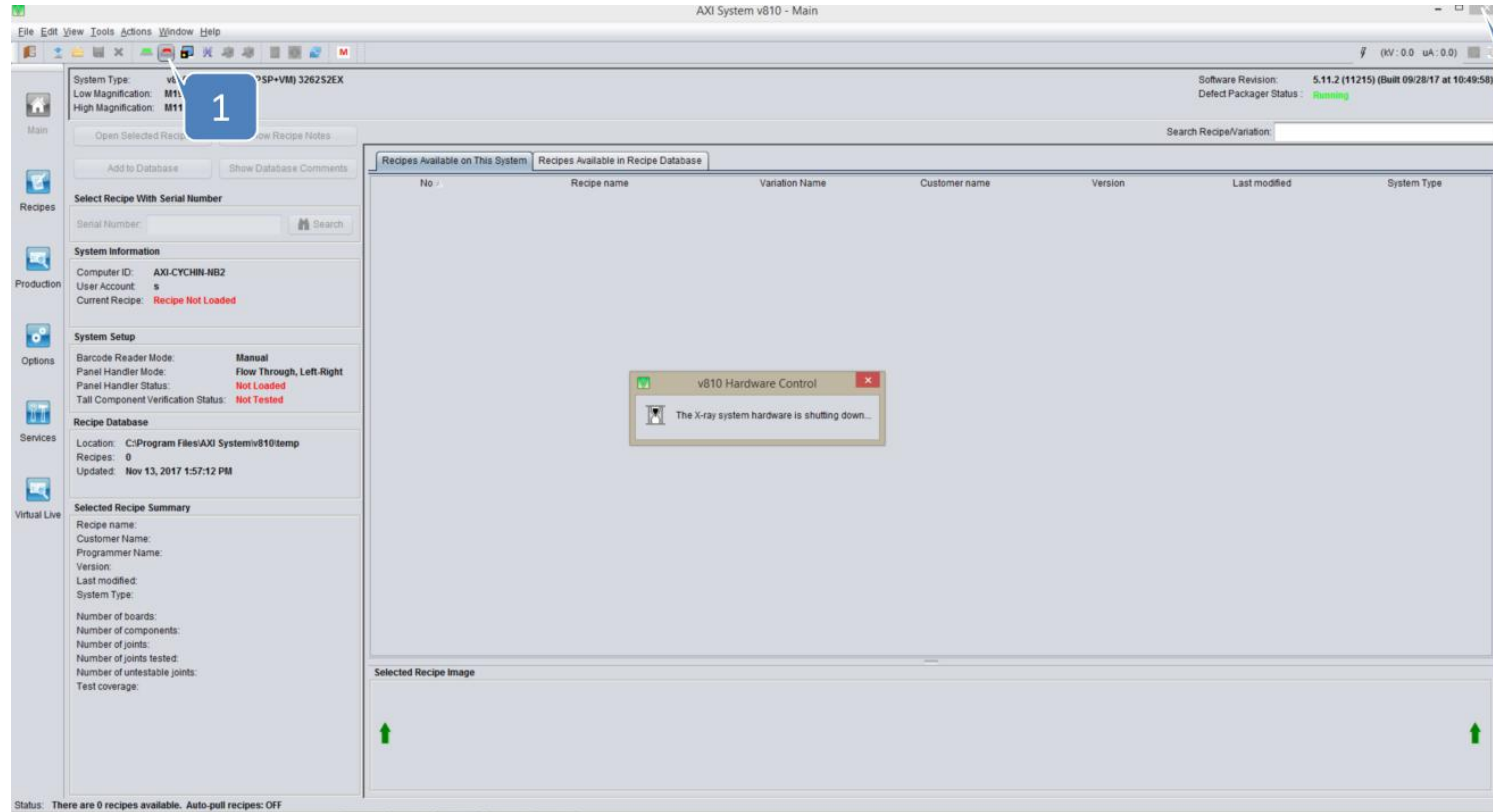
- At main page , click Unload Icon to unload panel from system before machine shutdown.
(this is importance to avoid board crush or system error during next start-up)

System ShutDown - Whole System Shut down



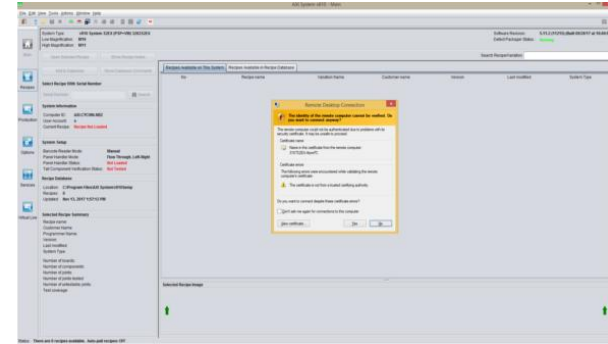
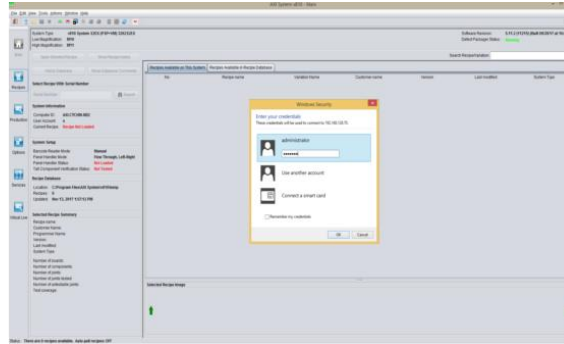
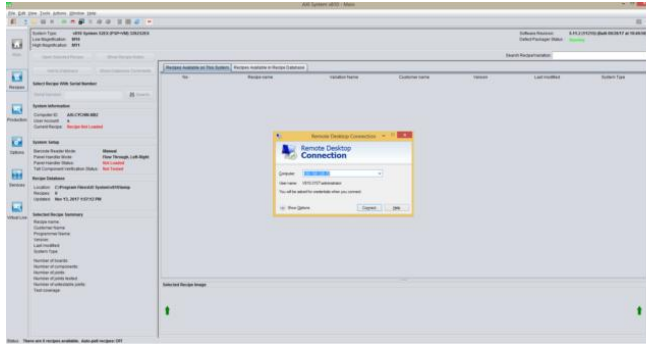
1. Click “Shut Down whole system” icon to completely power off machine including Atom PC. Remark, this feature only available for software version 5.11.2 and above.
2. Click “Yes” to continues and system will force shutdown main controller and RMS (Atom PC) without save other running task in progress.
3. Once done, may proceed turn off Xray source at Operator Console, main power input, SSCA power, press EMO and turn off air pressure inlet

System ShutDown - GUI



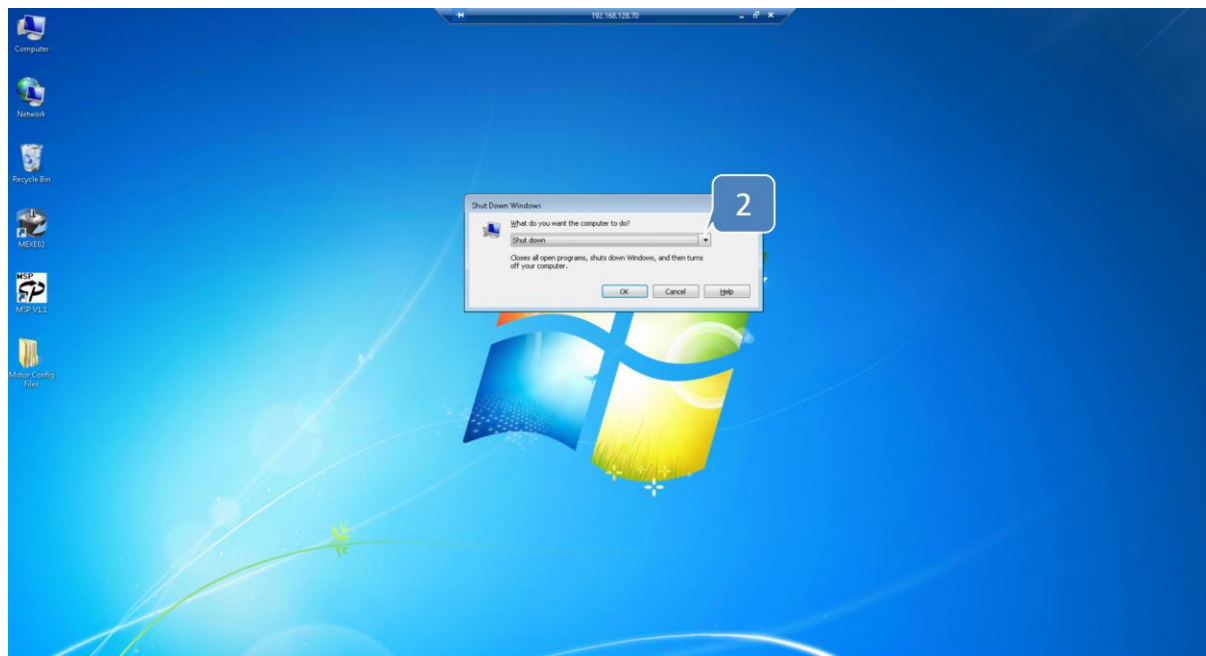
1. Click “Shut Down” icon to turn off X-ray sources and deactivate system hardware peripheral.
2. Exit the V810 GUI

System ShutDown - RMS (Atom PC)



1. Launch “Remote Desktop Connection” from main controller and remote to Atom PC by using IP “192.168.128.70”
2. Log in with user “Administrator” & Password “Please!”
3. Click “Yes” to proceed

System ShutDown - RMS (Atom PC)



1. Press “Ctrl+F4” and action tab will shown
2. Select “Shutdown” and press “OK”, Atom PC will be automatically shutdown after few second.
3. Back to main controller, shut down PC as usual
4. Once done, may proceed turn off Xray source at Operator Console, main power input, SSCA power, press EMO and turn off air pressure inlet



WEE KAH KHIM
Vice President & General Manager

GARY LEONG
Sales Director

周道南
Sales General Manager
(China & Taiwan)

SEOW ZI YANG
Business Development
Manager

FAHMI HELMI
Business Development
Manager